Inverse Airfoil Design as Constrained Root-Finding: A Monolithic CST–Newton Formulation

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August 6, 2026

Abstract

Inverse airfoil design, recovering a geometry that produces a prescribed surface pressure or edge-velocity distribution, is recast here as a single determined nonlinear root-find rather than an objective-function search. Parameterising the surface with Class-Shape-Transformation (CST) coefficients that enter the geometry *linearly* makes the geometric Jacobian block exact, constant, and design-independent. It also makes geometric design constraints (leading-edge radius, trailing-edge thickness, inscribed area) linear algebraic rows rather than nonlinear predicates. Appending the CST coefficients as unknowns to a coupled viscous/inviscid Newton solver (mfoil) therefore converts constrained shape *optimisation* into constrained *root-finding*: one square system, no outer loop, no surrogate. The architecture is validated with a falsifiable self-consistency test: a known CST coefficient vector is recovered from its own self-generated target to $\|A - A^*\|_\infty = 2.75 \times 10^{-11}$ in 6 Newton iterations, with a visible quadratic tail. The recovered geometry reproduces the target in fully released (natural-transition) direct mode to $\Delta c_l = 3.4 \times 10^{-12}$. An ablation matrix identifies the primary uniqueness guard for the resulting square system: sensitivity-optimal (QR-pivoted) selection of target stations, not initial-guess quality, separates recovery of the true design from clean convergence to a spurious but residual-zeroing root. Measured against a competently-tuned nested Levenberg–Marquardt baseline under two independent fair-paired controls, the monolithic architecture requires 3.1–8.1x fewer flow-residual evaluations. This is a real but modest reduction, not the two-to-three-orders-of-magnitude headline hypothesised a priori, and it comes with determinism, guaranteed local quadratic convergence, exact analytic constraint-row Jacobians, and per-iteration failure-mode diagnostics that the nested baseline has no analogue for. A subsequent 20-section NACA panel evaluation, run at the winning configuration the ablation matrix identifies, recovers all 18 generable sections to err_free_inf $\leq 1.51 \times 10^{-10}$. This result is reported alongside the interval-arithmetic reason its formal pre-registered Wilson-confidence criterion still fails at this panel size. The architecture is further placed on a comparison table against MISES's own Modal-Inverse mode, the nearest prior CST-based inverse method, and the current generative/learned-surrogate inverse-design literature. The comparison is made on formulation class, cost, determinism, and constraint-handling grounds rather than a single flow-solve number, since the methods are not commensurable on that axis alone.

Keywords: inverse design; airfoil; class-shape transformation; Newton method; viscous–inviscid coupling; identifiability; constrained root-finding

1 Introduction

Inverse aerodynamic design, given a target surface pressure or isentropic Mach distribution, recovering the geometry that produces it, is classically posed either as a well-posed but restrictive conformal-mapping problem [Lighthill, 1945, Volpe and Melnik, 1986], or as an outer optimisation loop wrapped around a converged forward flow solve. The latter is the practical default in industry: a geometry parameterisation, a surrogate or gradient-based optimiser, and hundreds to thousands of fully converged flow solves per design. The motivation for the present paper is autobiographical. The author's own doctoral work on sCO_2 axial-turbine blading [Sathish et al., 2019] used exactly this pattern: Class-Shape-Transformation (CST) geometry, a B-spline control-point wrapper, and a Dakota/Kriging surrogate loop against MISES. That work already used CST for blade parameterisation, together with geometric constraints encoding expert design practice, but it used them *inside* an optimiser. The present paper's claim is that the linearity which made those constraints convenient there is strong enough to remove the optimiser altogether. The earlier future-work section named the open problem verbatim: integrating CST directly with a Newton-based flow solver to escape the nested loop ([design note Sathish, 2026, §1.4]).

That escape route already exists in one solver. MISES's Modal-Inverse mode [Drela, 1990] appends mode-shape amplitudes directly to its global Newton system and drives them so that computed surface speed matches a target: a determined root-find, not a search. This is not new; Drela described it in 1986–1990. What MISES's built-in mode shapes are *not*, however, is a complete, geometrically meaningful, analytic design basis. They are heuristic bump functions, not a family whose coefficients double as leading-edge radius, trailing-edge thickness, or inscribed area. Kulfan's CST parameterisation [Kulfan, 2008] is such a basis, and Masters et al. [Masters et al., 2017] showed it is among the most efficient analytic families across a 2000-airfoil comparison. CST has also already been coupled to inverse design once, by Lane & Marshall [Lane and Marshall, 2010]. There, CST is used to *smooth* a pressure-residual-driven normal-direction shape correction inside an outer iterative loop; CST is the smoother, not an unknown in a Newton system. The present work does not repeat either of these. It combines a property of CST that neither prior use exploits.

1.1 Related work

Classical inverse design. Inverse aerofoil design predates any of the parameterisations discussed below. Lighthill's conformal-mapping method [Lighthill, 1945] is the founding well-posed formulation: a target speed distribution on the unit circle is mapped conformally to the physical aerofoil. Three integral *closure conditions*, closure of the contour, a fixed leading-edge radius, and a prescribed circulation, are required for a solvable, self-consistent shape to exist at all. This closure-condition structure is the direct conceptual ancestor of §2.2's linear rows: both are answers to the same question, "what makes an arbitrarily chosen target realisable by *some* member of the design family," asked eighty years apart under different geometric representations. Volpe & Melnik [Volpe and Melnik, 1986] sharpened well-posedness for the transonic regime, showing that an inverse problem posed with pressure (rather than speed) as the target and with the wrong closure conditions is generically *ill*-posed. Existence and uniqueness are not automatic once the target is chosen independently of the geometry family, a caveat that motivates the identifiability discussion of §5.1 in a different (discrete, finite-dimensional) setting. Eppler's design method [Eppler and Somers, 1980] and Selig & Maughmer's PROFOIL [Selig and Maughmer, 1992] are the mature, still-used engineering descendants. Both are multi-point conformal-mapping inverse solvers that target several operating conditions' pressure distributions simultaneously by blending mode functions, and both remain outside the Newton-coupled, viscous-inclusive family this paper and MISES belong to. They close the inverse problem analytically at the potential-flow level, then couple to a boundary-layer method non-simultaneously, rather than solving flow and geometry as one system.

Parameterisation comparison. Sobieczky's PARSEC [Sobieczky, 1999] parameterises an aerofoil by twelve directly-meaningful quantities (LE radius, thickness/camber extrema and their locations, TE angle and thickness) fitted by a polynomial basis. It is geometrically transparent in the same spirit as CST's endpoint identities (Eq. (6)), but through a basis whose coefficients are not independent design freedoms in the same linear sense: PARSEC's basis functions depend on the target values themselves (the fit is re-solved per target), whereas CST's Bernstein basis $S_i(\psi)$ is fixed once n is chosen, independent of what shape is being represented. This is the property Eq. (5) depends on. Masters et al.'s benchmark [Masters et al., 2017] is the field's reference comparison: over 2000 aerofoils and several parameterisation families (CST, PARSEC, B-splines, Bezier, orthogonal/SVD/POD modes derived from a training database), CST is found to be among the most geometrically efficient *analytic* (i.e. not data-driven) families, needing roughly 10–14 coefficients per surface for sub-0.1%-chord fitting error on typical sections. Purely data-driven SVD/POD modes fit an existing database more efficiently still, at the cost of needing that database and losing CST's closed-form geometric-functional structure (§2.2) entirely. This paper's choice of CST is therefore not merely convenience. It is the parameterisation Masters et al.'s own comparison identifies as the best-supported analytic basis, and the one whose particular algebraic property (linearity in A) this paper's contribution depends on.

Newton-coupled viscous/inviscid solvers. The architecture this paper appends CST to is Drela & Giles's ISES lineage [Drela and Giles, 1987]: a two-equation integral boundary-layer closure coupled *simultaneously* (not iteratively) to a panel-method inviscid solution through a single global Newton system, with the e^n transition model providing the closure that makes laminar-to-turbulent transition a smooth part of that system rather than a switched boundary condition. mfoil [Fidkowski, 2022] is a modern, open, from-scratch reimplementation of exactly this architecture: the ISES/XFOIL viscous-inviscid coupling without XFOIL's Fortran legacy code, in Python, with an exposed sparse global Jacobian. This is why it serves as the Stage 1 testbed here (§2): the extended system of §2.3 could not be built without an analysis code that already exposes its own $\partial R/\partial U$ block for direct augmentation. Section 5.5 of the dossier ([design note Sathish, 2026]) additionally documents that MISES's Parametric-Inverse mode was designed from the outset to accept an arbitrary geometry-definition system via a black-box interface (`BPREAD`/`BPWRIT`/`BLDGEN`). The mechanism this paper's Stage 3 (§7) would use already exists in the target solver, and has never been exercised with a CST-as-unknowns formulation.

Modern learned inverse-design methods. A distinct, much newer line of work learns the pressure-to-geometry map directly from data rather than solving a physics-based system at all. Conditional GANs [Yilmaz and German, 2020] and, more recently, diffusion models learn a stochastic generative map from a target Cp (or aerodynamic-coefficient) vector to a geometry, trained offline on a large precomputed database, with inference reduced to a single (or few-step) forward network pass. A classifier-free-guided denoising diffusion model reports a 34.4% precision improvement over WGAN baselines on an airfoil-generation benchmark [Yu et al., 2025]. A hybrid approach, LF-PINN [Wang et al., 2024], trains a physics-informed surrogate directly on mfoil's own low-fidelity flowfields, the same solver used here, and reports at least an order-of-magnitude online inference speedup relative to running mfoil directly, again after an offline training investment. These methods answer a different question well: given a large library of prior designs and target statistics resembling that library, produce a plausible geometry near-instantly. They do not provide a convergence guarantee, an exact constraint-row Jacobian, or a certificate that the returned geometry actually reproduces the requested Cp to any specified tolerance. The network's output is evaluated *after* the fact by running a flow solver on it, not constrained *during* generation the way Eq. (12)'s equality rows are. §5.10's comparison table places this paper's architecture and the learned-inverse family on common columns (flow solves per design, determinism, constraint handling) explicitly, because the two are not competing solutions to the same technical problem so much as different points on an accuracy-guarantee vs. amortised-cost trade surface.

Turbomachinery inverse design. Lavagnoli [Lavagnoli, 2026] benchmarks Kolmogorov–Arnold Networks against MLP and Gaussian-process regressors for inverse turbine-passage design, trained on ≈30,000 blade profiles generated by an automated MISES-coupled optimisation pipeline (inlet flow angle $-50°$ to $0°$, outlet $50°$–$75°$). The trained network infers a loading-consistent geometry in ≈0.1 s with a mean outlet-angle error of $0.086°$. Structurally this is the same learned-surrogate pattern as the previous paragraph, specialised to cascades, and it is the closest published turbomachinery analogue of this paper's motivating problem (§1). It is, however, once more amortised-cost statistical inference over a precomputed design database, not a per-target root-find with a convergence certificate. Lavagnoli's pipeline includes a predicted-RMS *feasibility proxy* that flags statistically ill-posed inverse targets before training-set generation. This is the statistical-learning analogue of this paper's realisability guard (§2.7), independently motivated by the same underlying problem (some targets are simply not achievable by the design family in use), but answered by a learned classifier rather than a closed-form linearisation.

The precise novelty claim. CST's surface parameterisation

$$\zeta(\psi) = C(\psi) \sum_i A_i S_i(\psi) + \psi\, \zeta_T \tag{1}$$

is *linear* in the coefficient vector A. Two consequences follow, and neither is available to a generic (e.g. B-spline-with-wrapper) geometry representation once local-frame perturbations, LE tangent-angle displacements, or re-fitting are introduced to reach the same design freedom ([design note Sathish, 2026, §1.5]). (1) The geometric Jacobian block $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ is exact, constant, and design-independent: it is assembled once and never re-derived inside a Newton loop. (2) The geometric functionals that matter for manufacturability (LE radius, TE thickness, wedge angle, inscribed area, even leading-order curvature continuity) are *linear* in A, so they enter the Newton system as constant-coefficient algebraic rows rather than non-smooth predicates or penalty terms. The contribution of this paper is **not** "geometry unknowns in a Newton system" (that is Drela, 1986/1990), and **not** "CST used somewhere inside inverse design" (that is Lane & Marshall, 2010). It is the combination: an analytic basis whose linearity makes the geometric Jacobian exact and constant, *and* whose geometric functionals make design constraints linear rows, together converting *constrained shape optimisation* into *constrained root-finding*, one square Newton system, solved once, with no outer loop.

The corrected headline. A priori, the architecture was hypothesised to require "two to three orders of magnitude fewer flow solves, plus determinism and uniqueness" relative to a nested-optimisation baseline ([design note Sathish, 2026, §1.3]). That number was never measured against a competent baseline, and it is not repeated here. A controlled ablation study (§5) instead measures the architecture against a modern, properly-scaled, tightly-tolerance Levenberg–Marquardt solver [Virtanen et al., 2020]: **3–8x fewer flow-residual evaluations, depending on initialisation strategy**, under two independent fair-paired controls. This measured ratio comes with qualitative properties the flow-solve count does not capture: determinism (no stochastic restarts), guaranteed local quadratic convergence, exact analytic constraint-row Jacobians (no finite-difference noise in the constraint block), and a battery of per-iteration failure-mode diagnostics (§3) the nested baseline has no equivalent of. The paper's two central empirical results are therefore the correction of the a priori ≥100x claim and the discovery of a station-selection identifiability mechanism that the original hypothesis did not anticipate (§5.1).

The remainder of the paper: §2 states the CST mathematics, the constraint rows, the extended Newton system and its degree- of-freedom accounting, the derivative strategy actually forced by the target solver's exposed interface, and the two solver-specific closures (forced transition, pre-solve/realisability) needed to make the Newton iteration well-posed. §3 catalogues the four failure modes the architecture is vulnerable to and the diagnostics built to discriminate between them. §4 describes the self-consistency test that falsifies or confirms the hypothesis outright. §5 reports the T8 ablation matrix: the identifiability finding, the Bernstein-conditioning cliff, the corrected flow-solve comparison, and the quadratic-convergence evidence. §6 places the four solver-specific mitigations in a common TRIZ separation-principle frame, states the study's scope limits, and discusses the outdated-baseline finding. §7 closes with the Stage 2 (cascade) and Stage 3 (MISES) extensions this architecture was designed to scale into.

2 Formulation

2.1 CST parameterisation and its linearity

Let $\psi = x/c \in [0, 1]$ and $\zeta = z/c$. The CST class function for a round-nose, sharp-tail family is

$$C_{N_1}^{N_2}(\psi) = \psi^{N_1}(1-\psi)^{N_2}, \qquad N_1 = 0.5,\ N_2 = 1.0, \tag{2}$$

and the shape function is a Bernstein-polynomial expansion of order n per surface,

$$S_i(\psi) = K_i\, \psi^i (1-\psi)^{n-i}, \quad K_i = \binom{n}{i}, \qquad S_u(\psi) = \sum_{i=0}^{n} A_{u,i} S_i(\psi), \quad S_l(\psi) = \sum_{i=0}^{n} A_{l,i} S_i(\psi). \tag{3}$$

The upper and lower surfaces are

$$\zeta_u(\psi) = C(\psi)\, S_u(\psi) + \psi\, \zeta_{T,u}, \qquad \zeta_l(\psi) = C(\psi)\, S_l(\psi) + \psi\, \zeta_{T,l}, \tag{4}$$

with $\zeta_{T,u} = +t_{TE}/2$, $\zeta_{T,l} = -t_{TE}/2$. The sign convention is binding throughout: lower-surface coefficients are stored with their natural negative sign, so the shared-leading-edge row reads $A_{u,0} + A_{l,0} = 0$.

Figure 1 shows the construction term by term. The class function of panel (a) is fixed geometry that carries no dependence on A at all; the Bernstein shape functions of panel (b) are fixed polynomials; and the surface of panel (d) is the A-weighted sum of the terms in panel (c). Only the choice of N_1 and N_2 is nonlinear, and those are held fixed throughout this work at the round-nosed, sharp-trailing-edge values $N_1 = \frac{1}{2}$, $N_2 = 1$.

The property this paper exploits is that Eq. (4) is *linear* in A: the partial derivative

$$\frac{\partial \zeta}{\partial A_i} = C(\psi)\, S_i(\psi) \tag{5}$$

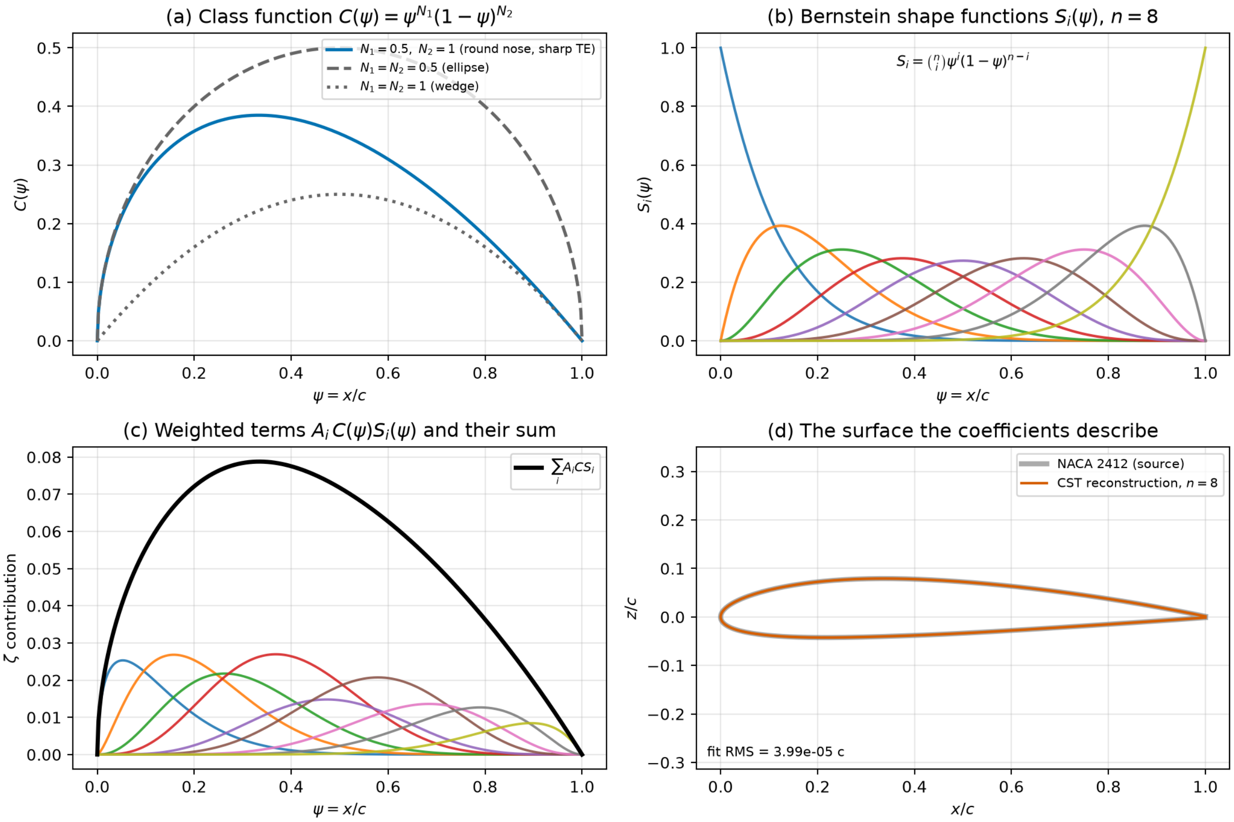

Figure 1: The CST construction, evaluated rather than sketched. (a) the class function for three (N_1, N_2) pairs, with the airfoil-like choice used here highlighted; (b) the Bernstein shape functions at $n = 8$; (c) the weighted terms $A_i C(\psi) S_i(\psi)$ for a fitted NACA 2412 upper surface, and their sum; (d) the resulting surface against the section it was fitted to. Every panel except the class function is linear in A, which is the property Eq. (5) states and the rest of the formulation depends on.

does not depend on A. It is a fixed matrix, computed once from the chordwise node distribution and cached (T2 requirement, [design note Sathish, 2026, §7.3]): *"does not depend on A. Recomputing it per Newton iteration is the single most common way to lose the performance argument."* This design-independence is what makes the $\partial R/\partial A$ Jacobian block of §2.3 assemble as a cheap matrix product rather than a re-derived, geometry- dependent operator at every Newton step.

Two endpoint identities follow directly from Eq. (4) and are used as constraint rows in §2.2. The trailing-edge identity is $\zeta'(1) = \zeta_T - A_n$ at $N_2 = 1$, and it does *not* take the same form on both surfaces: with the boat-tail half-angle defined per surface as the wedge closing angle, the upper surface descends toward the trailing edge ($\tan\beta_u = -\zeta_u'(1)$) while the lower surface rises ($\tan\beta_l = +\zeta_l'(1)$), so

$$A_0 = \sqrt{2R_{LE}/c}, \qquad A_{u,n} = \zeta_{T,u}/c + \tan\beta_u, \qquad A_{l,n} = \zeta_{T,l}/c - \tan\beta_l. \tag{6}$$

The sign flip on the lower surface is stated explicitly because applying the upper-surface form to a lower-surface coefficient, which is negative under the stored-sign convention, silently produces the wrong constraint row; that error occurred once during development and is now pinned by a test against the fitted trailing-edge slope. Near $\psi \to 0$, $\zeta \approx A_0\sqrt{\psi}$, so $R_{LE} = A_0^2/2$; G^2 curvature continuity across the leading edge reduces to matching the leading coefficient magnitude on the two surfaces, $A_{u,0} = |A_{l,0}|$; under the stored-sign convention ($A_{l,i} < 0$), the row $A_{u,0} + A_{l,0} = 0$.

Figure 2 gives these identities their physical reading and checks the design-independence claim numerically. Panels (a) and (b) plot the two endpoint identities of Eq. (6) with the fitted NACA 2412 values marked, so the coefficients that the inverse solve returns can be read directly as a nose radius and a trailing-edge wedge angle rather than as abstract weights. Panel (c) is the one that matters for the Newton system: the markers are the surface displacement measured by actually perturbing one coefficient of the fitted section and re-evaluating the geometry, and the lines are the analytic prediction $\delta A_i\, CS_i$. They agree to the level printed on the panel, and the agreement does not depend on where the remaining coefficients sit, which is exactly what permits $\partial\zeta/\partial A$ to be assembled once and cached.

2.2 Constraints as linear rows

Because Eq. (4) is linear in A, every geometric functional that is itself linear or integral in ζ becomes a linear row $g \cdot A = b$ rather than a nonlinear predicate evaluated on a rebuilt mesh. The leading-edge-radius and trailing-edge-wedge rows follow immediately from Eq. (6). The one requiring a closed-form integral is inscribed area: the per-surface contribution is

$$\int_0^1 C(\psi)S_i(\psi)\,d\psi = K_i \int_0^1 \psi^{i+N_1}(1-\psi)^{n-i+N_2}\,d\psi = K_i\, B(i+N_1+1,\, n-i+N_2+1), \tag{7}$$

a Beta function, plus the trailing-edge term $\int_0^1 \psi\,\zeta_T\,d\psi = \zeta_T/2$. Eq. (7) is evaluated with `scipy.special.beta`, never by numerical quadrature in a production constraint row; quadrature is retained only as an independent gate check. That gate passed: `area_row` matches numerical quadrature of the fitted NACA 2412 to 10^{-10}. This closed-form machinery is the same Beta-function apparatus Joseph & Mohan [Joseph and Mohan, 2021] use for thin-airfoil lift/moment integrals against a Bernstein camber basis. This is not a coincidence: both integrate Bernstein polynomials against the class function, and it is the analytic kernel reused for the linear pre-solve of §2.7.

Under a B-spline-with-wrapper representation, by contrast, every row in Table 1 is a nonlinear or non-smooth predicate evaluated on a rebuilt mesh. This is the stack that forced surrogate-based search in the author's earlier turbine-blading work [Sathish et al., 2019] ([design note Sathish, 2026, §1.5]).

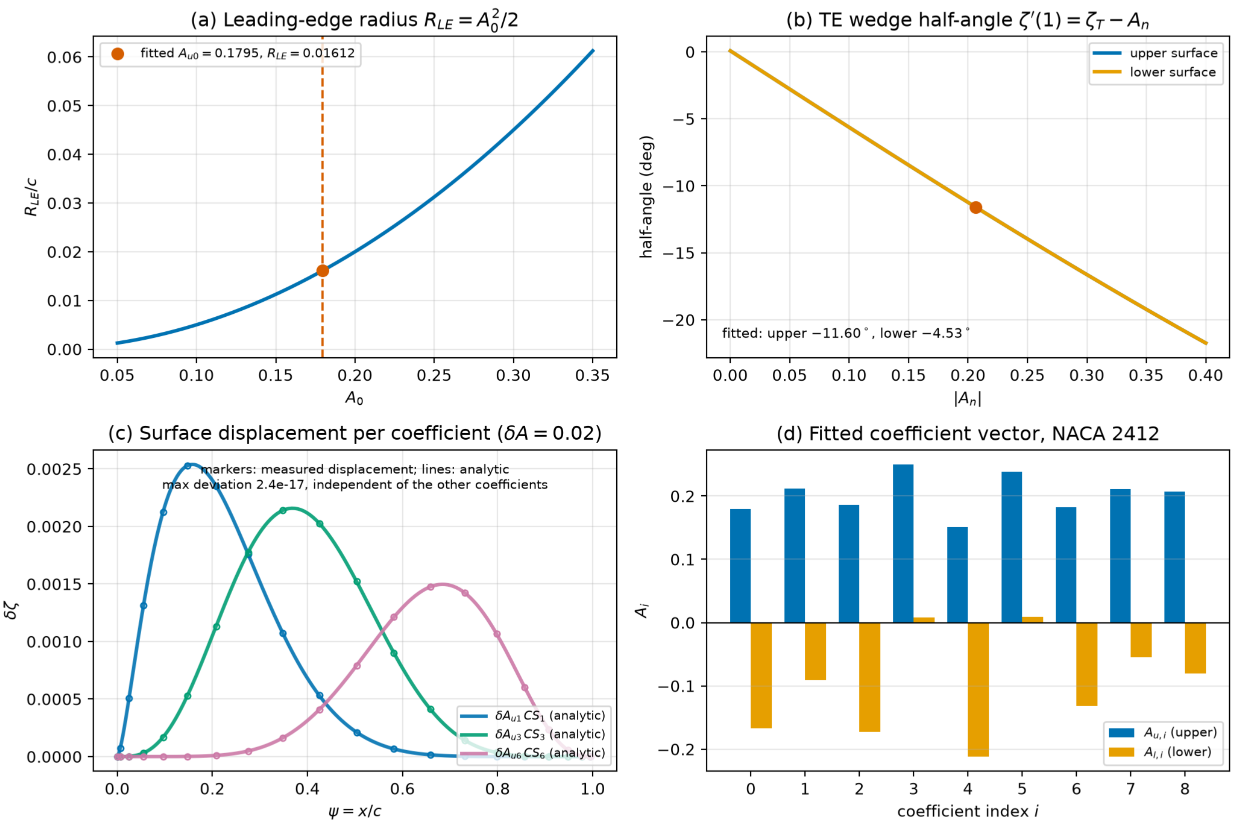

Figure 2: What the coefficients mean, and why the geometric Jacobian block is constant. (a) leading-edge radius against A_0, from Eq. (6), with the fitted NACA 2412 value marked; (b) trailing-edge wedge half-angle against the final coefficient, for both surfaces under the stored-sign convention; (c) superposition checked rather than asserted, markers measured by perturbing a single coefficient and lines predicted analytically; (d) the fitted coefficient vector itself, upper and lower blocks, in the ordering used throughout (§2.3).

Which rows the reported results actually exercise. The constraint machinery described here is broader than what the experiments in §5 use, and the difference should be stated rather than left to be inferred. Every configuration reported in this paper runs either with the leading edge prescribed, in which case $K = 0$ and no constraint row is assembled at all, or with `le_treatment=none`, in which case the single shared-leading-edge row $A_{u,0} + A_{l,0} = 0$ is used. The trailing-edge-wedge and inscribed-area rows, including the Beta-function derivation of Appendix A, are implemented and unit-tested against independent quadrature but are not exercised by any result reported here. They are presented as part of the formulation because they follow from the same linearity and cost nothing to assemble, not as validated contributions to the measured outcomes.

Table 1: Constraint representation under CST against a B-spline control-point wrapper. Every row that is a linear, exact algebraic condition on the CST coefficients becomes a nonlinear or non-smooth predicate on a rebuilt mesh under the wrapper, which is what forces an outer search rather than a root-find. Source: [design note Sathish, 2026, §1.5].

Constraint	Under CST	Under the B-spline wrapper
LE radius	$A_0 = \sqrt{2R_{LE}/c}$: linear, exact	implicit, via tangent points
TE thickness	ζ_T term: linear	geometric post-check
Boat-tail/wedge angle	A_n: linear	nonlinear angular DV
Inscribed area	linear functional, Beta-function closed form	numerical, non-smooth
Curvature	rational function of A, differentiable	monotonicity predicate, non-smooth

2.3 The extended Newton system and DOF accounting

The extended system appends the CST coefficients (and, optionally, angle of attack) directly to the flow solver's own global unknown vector. Unknowns:

$$U \text{ (flow state}, N_{tot}), \qquad A \text{ (CST coefficients}, n_A), \qquad \alpha \text{ (angle of attack}, 0 \text{ or } 1). \tag{8}$$

Equations:

$$R(U; x(A)) = 0 \;\; (N_{tot}), \qquad T(U) - Cp_{target} = 0 \;\; (M), \qquad G \cdot A - b = 0 \;\; (K), \tag{9}$$

with the block Jacobian, dimensions annotated below each block for the T7 winning configuration. There, mfoil's global system carries $N_{sys} = 230$ stations (200 airfoil-surface nodes plus wake, §4) with four boundary-layer state variables each, so $N_{tot} = 4N_{sys} = 920$; the prescribed-LE treatment leaves $n_{A,\text{free}} = 16$ of the 18 CST coefficients as unknowns; and $M = 16$, $K = 0$, $n_\alpha = 0$. The assembled system is therefore $J \in \mathbb{R}^{936\times936}$, and the recorded rank at the first iteration is 936, that is, full (`experiments/results/t7_naca2412/diagnostics.json`):

$$J = \left[\begin{array}{c|c|c} \underbrace{\partial R/\partial U}_{N_{tot}\times N_{tot}} & \underbrace{\partial R/\partial A}_{N_{tot}\times n_{A,\text{free}}} & \underbrace{\partial R/\partial \alpha}_{N_{tot}\times n_\alpha} \\ \hline \underbrace{\partial T/\partial U}_{M\times N_{tot}} & \underbrace{0}_{M\times n_{A,\text{free}}} & \underbrace{\partial T/\partial \alpha}_{M\times n_\alpha} \\ \hline \underbrace{0}_{K\times N_{tot}} & \underbrace{\partial G/\partial A}_{K\times n_{A,\text{free}}} & \underbrace{0}_{K\times n_\alpha} \end{array}\right] \in \mathbb{R}^{(N_{tot}+M+K)\times(N_{tot}+n_{A,\text{free}}+n_\alpha)}, \tag{10}$$

square by Eq. (11)'s construction-time assertion $(M + K = n_{A,\text{free}} + n_\alpha)$, so the $M + K$ extra rows exactly balance the $n_{A,\text{free}} + n_\alpha$ extra columns beyond the $N_{tot} \times N_{tot}$ flow block, which is itself always square. The $M \times n_{A,\text{free}}$ zero block in the middle column reflects that the target-Cp equations $T(U) - Cp_{target}$ depend on A only *through* U (Cp is read off the converged flow state, not directly from geometry). Its entire A-sensitivity is therefore carried through the $\partial R/\partial A$ column and the chain of Newton updates; there is no direct T-vs-A coupling to assemble. $\partial R/\partial U$ and $\partial R/\partial \alpha$ are mfoil's own analytic, sparsely-assembled blocks, reused unchanged. $\partial T/\partial U$ is a near-selection matrix (Cp at a node as a direct function of the panel/edge-velocity unknowns). $\partial G/\partial A = G$ is constant and free, per §2.2. $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ is the block that carries the architectural claim; its second factor is the cached, design-independent matrix of Eq. (5), and its first factor is addressed in §2.4.

DOF accounting. The dossier's original bookkeeping ([design note Sathish, 2026, §4] FM-1, §7.6) is

$$\text{DOF} = 2(n+1) - 1_{\text{shared LE}} - 1_{\text{TE fixed}} + 3_{\text{scale/rotate/translate}} + 1_{\text{stagger or } \alpha} \quad \overset{!}{=} \quad M + K.$$

In the Stage-1 (isolated-airfoil, mfoil) implementation, the three rigid overall modes (scale/rotate/translate) are not exposed as free Newton unknowns, so the implemented squareness condition is the simpler, equivalent statement actually asserted at construction time and logged every run:

$$M + K = n_{A,\text{free}} + n_\alpha, \tag{11}$$

where $n_{A,\text{free}}$ is the count of CST coefficients *not* eliminated by a prescribed-LE treatment (§4) or an equality-row elimination, and $n_\alpha \in \{0, 1\}$ depending on whether angle of attack is a free unknown for the run in question. Violating Eq. (11) raises at construction, before any Newton step. This is the FM-1 guard (§3), and it is exercised directly in the T8 ablation matrix (§5.1): deliberately over- and under- squaring the baseline system (n=8/side, $n_{A,\text{free}} = 16$, $n_\alpha = 0$) produces the exact, actionable diagnostics

```
"extended system not square: M+K=17 but
  n_A_free+n_alpha=16 (M=17, K=0, n_A_free=16)"
"extended system not square: M+K=15 but
  n_A_free+n_alpha=16 (M=15, K=0, n_A_free=16)"
```

with zero Newton iterations spent and no silent wrong answer. Getting Eq. (11) wrong is, per the dossier, plausibly what the author's PhD-era divergence was misdiagnosed as an LE-singularity problem when it was rank deficiency ([design note Sathish, 2026, §4], FM-1). Instrumenting for it directly, rather than inferring it from Newton behaviour, is the point of Eq. (11) being an assertion rather than a derived quantity.

2.4 Derivative strategy: why finite-difference-over-A, not the analytic chain rule or complex-step

The derivative-mode decision tree adopted for this work prefers, in order: (1) the analytic chain rule $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ if mfoil exposes $\partial R/\partial x$ analytically; (2) complex-step over A if the residual-assembly path is free of `abs`/`max`/`min`/comparison operators; (3) real (central) finite differences over A otherwise. T1 introspection resolved both questions empirically, and neither resolved in the analytic-chain-rule's favour:

- **The analytic chain rule is incomplete for this solver class.** mfoil exposes `M.glob.R_x`, but it is only a *partial* sensitivity: real and useful for the arc-length (ξ) mapping, but missing the geometry-dependent operators that matter most, the inviscid panel/AIC coupling and the stagnation-point formulation. Using this partial block as if it were the complete $\partial R/\partial x$ would silently drop exactly the rows the leading-edge failure mode (FM-3, §3) is about.

- **Complex-step is blocked, not merely inconvenient.** A bare `residual_station/residual_transition` call, with fixed panels and a complex-perturbed state, is verified complex-step-safe and reproduces the analytic R_x to float64 machine epsilon. The *full* augmented residual, which is what a perturbation in A actually varies since A moves the panel geometry itself, calls the panel-influence and stagnation-detection code paths. These use `atan2`, branch selection on real quantities, and a hard `ValueError` guard for complex node arrays. These are exactly the operators complex-step cannot tolerate, and they sit in precisely the geometry-dependent block that (1) shows was missing from the analytic path: complex-step *cannot fill the gap the analytic chain rule leaves.*

The implemented and gate-verified strategy is therefore real central finite differences with respect to A only, never with respect to node coordinates x (). This is a deliberate architectural discipline, not a fallback of convenience. A has $n_A \approx 16$–36 columns depending on Bernstein order, so a full central-difference Jacobian block costs $O(n_A)$ residual *evaluations* per Newton step, not flow solves, against the alternative of finite-differencing over $N_{panel} \gg n_A$ node coordinates. That alternative would silently destroy the flow-solve-count argument this paper's Results section depends on. T5 gate closure verified the resulting Jacobian columns to 5–6 significant figures across a step-size sweep with no stencil discontinuities. The step size itself (fd_step $= 10^{-6}$, `configs/default.yaml`) is not hand-tuned per case. The T5 sweep varied it across several decades and confirmed the resulting FD-column agreement with the (independently verified, §2.4 above) complex-step-safe partial block was flat across the sweep's central range, with no visible truncation/roundoff trade-off cliff at the scales exercised. That is, 10^{-6} sits well inside a broad plateau, not at a knife-edge optimum a different problem instance would move.

2.5 Algorithmic summary

Figure 3 gives the whole procedure in one view before the pseudocode states it precisely. The two shaded regions are the paper's two structural claims: everything left of the Newton loop is set up once and never revisited, and the loop itself solves a single square system rather than driving an outer optimiser. The two decision points are the only places the procedure can refuse to proceed, and both are diagnostics rather than tolerances: the squareness assertion of Eq. (11) is a statement about degrees of freedom, and the realisability gate is a statement about whether the target lies in the reachable set at all.

Algorithm 1 states the full monolithic iteration, target generation through release-and-verify, as pseudocode, making explicit where each of §2.3–§2.7's mechanisms (FD Jacobian assembly, joint under-relaxation, the trust region on ΔA, the forced-transition shim) sits inside the loop. It is a direct pseudocode rendering of `src/cins/solver/newton.py::solve_inverse` and `src/cins/benchmarks/pipeline.py::run_pipeline`, not an idealisation of it. Algorithm 2 states the QR-pivoted station-selection procedure identified as the identifiability guard in §5.1 (`src/cins/benchmarks/pipeline.py`, station- selection step), reusing the T4 pre-solve sensitivity matrix rather than building a second one.

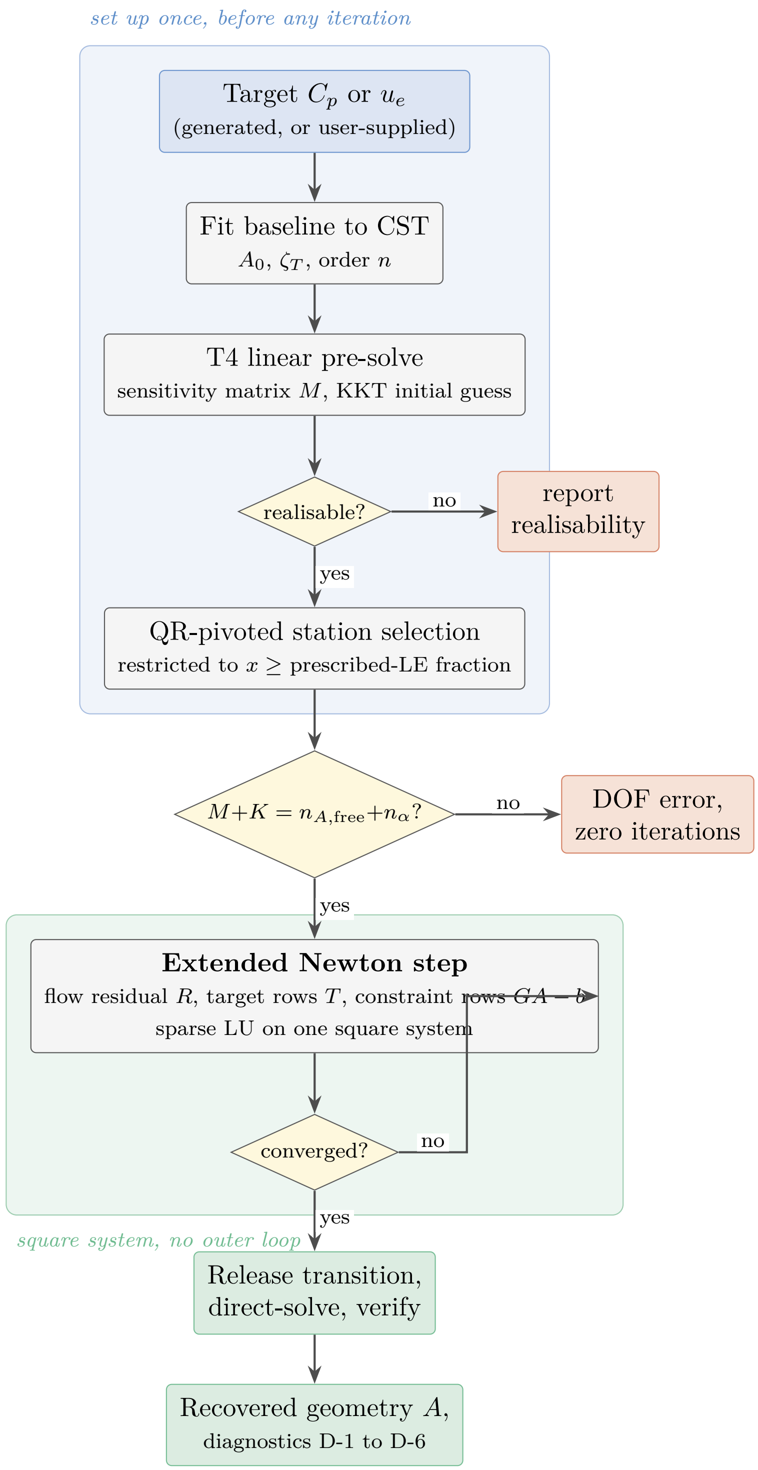

Figure 3: The monolithic inverse-design procedure. The pre-solve, station selection and degree-of-freedom check all happen once, before any Newton iteration; the loop then solves one square system per step with no outer optimiser. The realisability gate reports rather than blocks, because realisability is an inviscid-consistent quantity (§2.7) and a viscous target can be reachable while failing it. Algorithms 1 and 2 state the two shaded stages precisely.

Algorithm 1 Monolithic CST–Newton inverse iteration (§2.3–§2.7)

Require: target airfoil (for A^* generation) or externally supplied Cp_{target}; Bernstein order n; `le_treatment`; `transition.mode`

Fit CST to reference geometry $\rightarrow A^*$ (§2.1); build $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ once, cache (Eq. (5))
Generate target: direct-solve at A^* (forced transition if `mode=forced`, §2.6) $\rightarrow Cp^{\text{full}}_{target}$
Perturb or randomize $A^* \rightarrow A_0$ (T8 `init` factor); assemble G, b from §2.2
if `init = presolve` **then**
  for 2 passes **do** ▷ T4 pre-solve, §2.7
    Build sensitivity matrix M at current A_0 (finite-difference or Beta-function analytic)
    Solve KKT system Eq. (12) for A_0; record realisability, model_gap (ADR-0004, two *distinct* metrics)
  end for
end if
Select M target stations by QR pivoting (Algorithm 2) or evenly-spaced (T8 ablation)
Assert $M + K = n_{A,\text{free}} + n_\alpha$ (Eq. (11), FM-1 guard) ▷ raises before any Newton step if violated
Direct-solve flow at A_0 (converges U_0) ▷ initial point for the extended Newton loop
$k \leftarrow 0$
while $\|R\|, \|T\|, \|G\| >$ `rtol` and $k <$ `max_iter` **do** ▷ D-1 per-block residual norms
  Assemble $\partial R/\partial U$, $\partial R/\partial\alpha$ from mfoil (analytic, unchanged)
  Assemble $\partial R/\partial A$ by central finite differences over A only (§2.4), $O(n_A)$ residual evaluations
  Assemble $\partial T/\partial U$ (Cp-selection map), $\partial G/\partial A = G$ (constant)
  Solve linearised system $J\,\delta \leftarrow -[R; T - Cp_{target}; G \cdot A - b]$ (Eq. (10))
  Compute joint under-relaxation $\omega \in (0, 1]$: the smaller of mfoil's own U-limiter (Hk, θ/δ^* bounds) and the CST trust region $\|\delta_A\|_\infty \leq$ `a_trust_radius` (§2.3)
  $U \leftarrow U + \omega\,\delta_U$; $A \leftarrow A + \omega\,\delta_A$; $\alpha \leftarrow \alpha + \omega\,\delta_\alpha$ (if free)
  Rebuild geometry from updated A (apply_geometry, §2.1); record D-2..D-6 diagnostics
  $k \leftarrow k + 1$
end while
Release-and-verify (§2.6): un-force transition; direct-solve at converged A; assert $\Delta c_l, \Delta c_d$ against the natural-mode target within gate
return A, iteration count k, residual history, release-and-verify record

Algorithm 2 QR-pivoted target-station selection (§5.1)

Require: sensitivity matrix $M \in \mathbb{R}^{N_{cand} \times n_A}$ from the T4 pre-solve (§2.7); free-coefficient index set $\mathcal{F}$; required station count $M_{req} = n_{A,\text{free}} - K + n_\alpha$

Restrict candidates to nodes at or aft of the prescribed-LE fraction (FM-3 mitigation, §3)
Form the candidate submap $\tilde{M} \leftarrow M[\text{candidates}, \mathcal{F}]$
Column-pivoted QR: $\tilde{M}^\top P = QR$, giving a pivot ordering P by decreasing contribution to $\tilde{M}^\top$'s numerical rank
Select the first M_{req} pivoted candidate indices $\rightarrow$ target stations
return sorted station index set; report $\text{cond}(\tilde{M}[\text{stations}, :])$ (D-3-class diagnostic, *distinct* from $\text{cond}(J)$, ADR-0004)

2.6 Forced-transition onset closure

variables: the transition-station equation itself performs an internal sub-Newton solve for the point x_t where the marched amplification factor crosses n_{crit}, and that location moves discontinuously as geometry changes cross a threshold. A coupled Newton iteration needs a continuous residual; a kink stalls or chatters it (dossier §4, FM-4). The mitigation prescribed by the dossier is to fix (trip) transition during the inverse iterations and release it for a verification solve afterward. mfoil exposes no forced-trip API, so this is implemented as an adapter-level shim (ADR-0003) rather than a vendor-code edit.

The first implementation attempt froze the laminar/turbulent node flags (`M.vsol.turb`) and no-op'd mfoil's own `update_transition`, but left the natural `residual_transition` equation untouched at the frozen boundary station. This was *insufficient*. Empirically (NACA 2412, $\alpha = 2°$, $Re = 10^6$, trip 5%/5%), the physics moved correctly under Newton pressure (c_d: $0.00578 \rightarrow 0.00741$), but the solve never converged. `Rnorm` stalled at ≈ 12.6 for the full 50-iteration budget, because the frozen station equation still solves internally for amplification $= n_{crit}$, which has no solution once transition is pinned somewhere amplification is nowhere near n_{crit} (ADR-0003, "Revision"). The corrected mechanism, `_residual_transition_forced`, replaces the boundary-station residual with an XFOIL-style turbulence-*onset* closure: the momentum and shape-parameter rows use the ordinary turbulent `residual_station` closure directly across the whole station (no laminar sub-leg, no station-local x_t), and the shear-lag row is replaced with the same onset condition mfoil's own `update_transition` uses to initialize a newly-turbulent node, $U_2[2] = c_{ttr}(U_2)$, which depends only on the downstream state (ADR-0003). With this closure, the shim converges cleanly: trip 5%/5% reaches `glob.conv=True`, Rnorm $< 10^{-10}$, $c_d = 0.01125$ (strictly $>$ natural $c_d = 0.005778$, as physically expected for larger turbulent wetted area). Trip 30%/30% gives $c_d = 0.00788$, strictly between the 5%-trip and natural values, monotonic in trip location. Releasing transition and re-solving from the converged forced state reproduces the pinned natural baseline to within its stated tolerance ($c_l = 0.449351 \pm 10^{-3}$, $c_d = 0.005778 \pm 2 \times 10^{-4}$). These figures are from the shim's own verification run (ADR-0003) and differ in the fourth decimal from the T7 archive's release-and-verify values quoted in §4 ($c_l = 0.449091$, $c_d = 0.005774$), which is a different run at a different paneling, not a restatement of the same one. Rebuilding the global system after convergence with the shim still active reproduces $\|R\| < 10^{-8}$: the converged state is a genuine fixed point of the forced residual, not an artifact of the step sequence (ADR-0003, "Consequences"). Every T7/T8 result reported in §5 therefore states its transition mode explicitly, and release-and-verify in natural-transition direct mode is part of the T7 protocol (§4), not an optional extra.

Figure 4 shows the state this closure operates on, and is the reason a coupled solver was required rather than an inviscid one with a boundary-layer post-process. The inverse system of §2.3 is appended to mfoil's *viscous* global system, so the six distributions in the figure are unknowns inside the same Newton solve that recovers the geometry, not quantities computed afterwards from a converged pressure field. Panel (f) makes the coupling explicit: the displacement thickness δ^* displaces the effective body the inviscid half sees, and it is through that term that a geometry perturbation reaches the boundary layer and back. The trip locations are marked on each panel; the skin-friction rise at the trip in panel (c) and the corresponding drop in shape factor in panel (d) are the closure described above taking effect.

2.7 Analytic pre-solve and realisability

Two jobs are done by a single linear pre-solve before any Newton iteration runs, and the distinction between them is load-bearing for how results are reported (ADR-0004). Thin-airfoil theory is linear

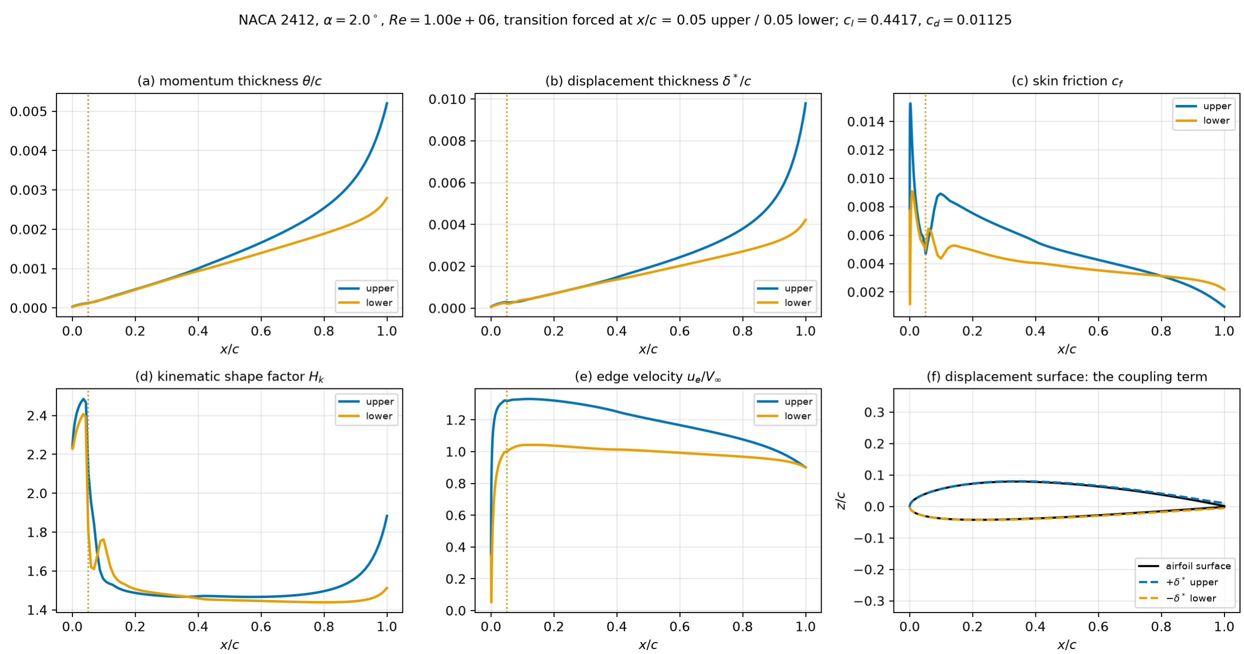

Figure 4: The coupled viscous solution the inverse system is embedded in, for NACA 2412 at $\alpha = 2°$, $\mathrm{Re} = 10^6$, with transition forced at 5 percent chord on both surfaces (dotted lines). (a)–(b) momentum and displacement thickness; (c) skin friction, rising at the trip; (d) kinematic shape factor; (e) edge velocity, the quantity the inviscid half exchanges with the boundary layer; (f) the displacement surface $\zeta \pm \delta^*$, which is the coupling term itself. All six are state variables of the same global system to which the CST coefficients are appended.

in camber slope and thickness, and CST camber/thickness are linear in A, so $Cp_{target} \approx M \cdot A$ for a sensitivity matrix M built either analytically (the Joseph–Mohan Beta-function kernels) or, as implemented, numerically from mfoil's own linear-vorticity panel influence coefficients at fixed geometry. Solving the constrained least-squares problem with the §2.2 rows as linear equality constraints,

$$\begin{bmatrix} M^\top M & G^\top \\ G & 0 \end{bmatrix} \begin{bmatrix} A \\ \lambda \end{bmatrix} = \begin{bmatrix} M^\top Cp_{target} \\ b \end{bmatrix}, \tag{12}$$

gives (1) an initial guess placing Newton inside its basin of attraction: the LE region is exactly where Newton wanders, and mfoil's own documentation warns the coupled solver can fail from a poor initial guess; and (2) a **realisability** metric, $\|MA^* - Cp_{target}\|/\|Cp_{target}\|$, that flags targets outside the CST-representable manifold *before* a Newton solve is attempted.

An adversarial T4/T5/T7 review found these two jobs' outputs had been conflated into one number, and ADR-0004 is the resulting decision, binding for every number reported in this paper:

> 1. **CST-representability** (the realisability guard, above) must be answered with *model-consistent* quantities: inviscid target against inviscid linearisation. On the T7 winning configuration this is 3.90×10^{-4}; the target is representable. 2. **Model gap**, how far the inviscid linearisation sits from the viscous physics the Newton solve actually faces, is a *separate*, diagnostic-only quantity, 0.110 on the same case. This is boundary-layer displacement effect, which no inviscid model captures. It is *not* target unrealisability; the subsequent monolithic solve converging to 2.75×10^{-11} (§4) is direct proof the target was realisable despite the nonzero model gap.

Reporting rule, enforced throughout this paper: realisability numbers are always labelled with their model ("inviscid-consistent"), and the conditioning of the QR-selected station submap (cond ≈ 90, §5.1) is never quoted as the conditioning of the full extended Newton system. These are different matrices answering different questions; conflating them was one of the review's confirmed findings.

3 Failure-mode instrumentation

Four failure modes were identified a priori ([design note Sathish, 2026, §4]) as the ways the architecture of §2 can fail without that failure being obviously attributable to its true cause. Table 2 summarises each, its diagnostic, and its mitigation; §2.3– §2.7 describe FM-1, FM-4's mechanism and FM-1's realisability guard in detail, and §5.1 reports the empirical FM-2 cliff.

The remaining diagnostics, D-1 (per-block residual norms, discriminates *which* block stalls) and D-6 (Newton convergence history, the headline figure confirming the quadratic-tail architectural claim), are not tied to a single failure mode; they are the general-purpose instruments used throughout §5. FM-3's fix targets *rows*, not columns: §2.1 showed $\partial\zeta/\partial A_i$ is perfectly finite as $\psi \to 0$ (it tends to zero), so the geometric columns are healthy. What blows up is the flow residual's *sensitivity* to geometry near the nose (surface normals, metric terms, the stagnation-point mapping all have unbounded derivatives there), so a handful of rows near the stagnation point acquire enormous magnitude and destroy the coupled matrix's condition number. This is a concrete instance of the point made in §2.1: CST's linearity keeps the columns exact, but does not by itself guarantee the rows are well-conditioned near a genuinely non-analytic geometric feature.

Table 2: The four failure modes identified a priori, each with the diagnostic that detects it and the mitigation adopted. All four were predicted before implementation and all four were observed; the instrumentation exists so that a failed solve reports which mode it hit rather than only that it failed. Source:, [design note Sathish, 2026, §4].

ID	Name	Mechanism	Diagnostic	Mitigation
FM-1	DOF/realisability	rank-deficient or over-determined	D-2 (rank/cond J)	Eq. (11) assertion; T4 realisability
FM-2	Bernstein conditioning	Gram matrix cond $\sim 4^n$	D-3 (cond($G^\top G$) vs n)	keep $n \lesssim$ 10–12/side; orthogonalise
FM-3	LE-singular Jacobian rows	$\sqrt{\psi}$ non-analytic at $\psi = 0$	D-4 (row-norm vs station)	row scaling; prescribed-LE (§4)
FM-4	C^0 viscous closure	transition-station kink	D-5 (transition history)	forced trip, release-and-verify (§2.6)

4 The falsifiable test

mfoil's Newton system yields a determined, quadratically convergent root-find, is falsifiable by a single self-consistency experiment with a *known* answer, guaranteed realisable by construction. The protocol fits CST to a known airfoil, runs mfoil in direct mode to generate a target Cp/edge-velocity distribution from that fit, perturbs the coefficients (or uses the §2.7 pre-solve) to obtain a starting guess, and asks whether the monolithic inverse Newton iteration recovers the original coefficient vector A^*. Any failure of this test isolates cleanly to one of the Table 2 failure modes rather than to target infeasibility. This is why this test, not a search over independently chosen targets, is the load-bearing evidence for the hypothesis ([design note Sathish, 2026, §7.8], §7.10).

Winning configuration. NACA 2412, $n = 8$ per side, `le_treatment=prescribed` (the first $\approx$5% chord is fixed geometrically rather than left inverse: the "mixed direct-inverse" mitigation for FM-3, the cheapest test of the whole hypothesis since it isolates the LE question from the architecture question), transition forced (§2.6), stations chosen by QR column pivoting (§5.1), initialized from the T4 pre-solve, α fixed.

Protocol, including release-and-verify. Success criterion: $\|A - A^*\|_\infty < 10^{-4}$ within ≤ 9 Newton iterations, with a visible quadratic tail in the D-6 convergence history. Because the inverse solve runs under the forced- transition closure of §2.6, the protocol additionally requires releasing transition and re-solving in fully natural (direct) mode at the recovered geometry, verifying that the forced-transition solve did not silently converge to an aerodynamically different answer than the natural-transition physics would report (ADR-0003 mandate).

Result: PASS, with large margin (Table 3). Measured against a 10^{-4}/9-iteration gate: $\|A - A^*\|_\infty = 2.747 \times 10^{-11}$ in 6 Newton iterations, with the target-block residual history $5.85 \times 10^{-4} \to 2.05 \times 10^{-5} \to 1.57 \times 10^{-7} \to 1.21 \times 10^{-8} \to 2.14 \times 10^{-10} \to 9.47 \times 10^{-12}$ showing the quadratic tail directly (Figure 5). Release-and-verify against a $\Delta c_l < 10^{-3}$, $\Delta c_d < 2 \times 10^{-4}$ gate gives $\Delta c_l = 3.4 \times 10^{-12}$, $\Delta c_d = 3.0 \times 10^{-14}$, both essentially at floating-point noise: the forced-transition solve and the natural-transition physics agree to machine precision at the recovered design ($c_l = 0.449091$, $c_d = 0.005774$ under both). Realisability (inviscid-consistent, ADR-0004 metric 1) is 3.9×10^{-4}; model gap (diagnostic only) is 0.110 (§2.7). Sixteen coefficients are *recovered* by the Newton iteration; the two leading-edge coefficients $A_{u,0}$ and $A_{l,0}$ are *prescribed* by the `le_treatment=prescribed` configuration and are not recovered. The error norms quoted above are over the 16 free coefficients ([Sathish, 2026]).

Table 3: T7 falsifiable test at the winning configuration: PASS on every criterion, each by several orders of magnitude. Run under the (surface, x) target-station addressing of §2.3, which is *not* the addressing used for the ablation matrix of §5. Source: `experiments/results/t7_naca2412/result.json`.

Criterion	Threshold	Measured
$\|A - A^*\|_\infty$ over 16 free coefficients	$< 10^{-4}$	$\mathbf{2.747 \times 10^{-11}}$
Newton iterations	≤ 9	**6**
Quadratic tail (D-6)	visible	$5.85 \times 10^{-4} \to 1.57 \times 10^{-7} \to 9.47 \times 10^{-12}$
Release-and-verify Δc_l	$< 10^{-3}$	$\mathbf{3.4 \times 10^{-12}}$
Release-and-verify Δc_d	$< 2 \times 10^{-4}$	$\mathbf{3.0 \times 10^{-14}}$

An independently-run six-vector adversarial review probed circularity, station-index correspondence, the pinned-coefficient framing, the absence (at the time) of release-and-verify, realisability semantics, and the submap/system conditioning category error; every confirmed finding is reflected in this paper's reporting discipline: the free-16 against all-18 split above, and ADR-0004's two-metric realisability rule. The station-index correspondence finding was closed differently from the others. It was originally answered with a numeric guard, asserting that the selection-time node and the target's own paneling agreed to $\max |\Delta x/c| < 2 \times 10^{-5}$. That guard has since been removed, because addressing a station by (surface, x/c) and interpolating the target's own C_p curve at that x makes the correspondence exact rather than approximate, so there is no longer a quantity for the guard to bound (§2.3).

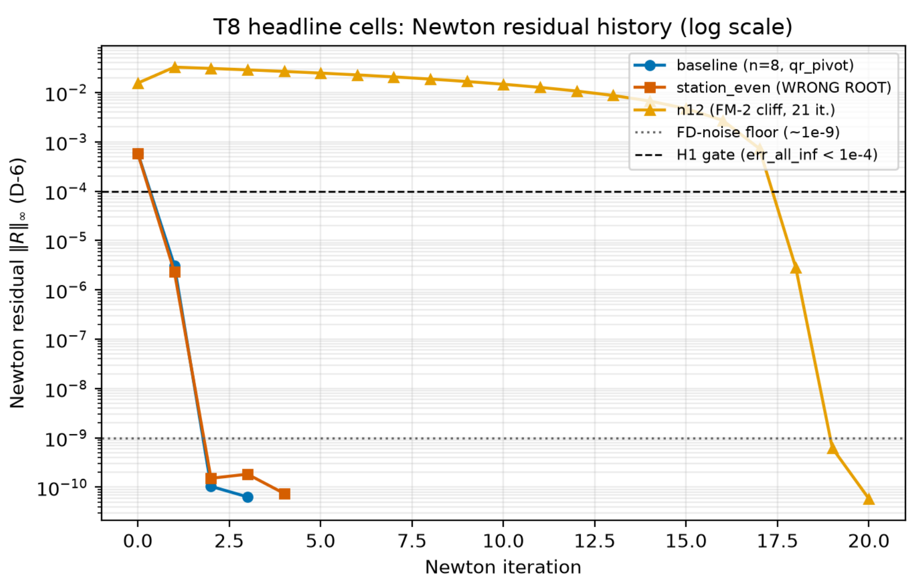

Figure 5: D-6: Newton residual-norm history for the T7 baseline, overlaid with the `t8_station_even` wrong-root case (§5.1) and the `t8_n12` conditioning-cliff case (§5.1). Source: `experiments/results/t8/figures/h3_convergence_overlay.png`.

The Newton residual history in Figure 5 is a scalar summary. Figures 6–8 give the geometry- and flow-field-level view of the same result, regenerated directly from a fresh re-run of the winning configuration (`src/cins/benchmarks/paper_figures.py`, never hand-plotted). Figure 6 overlays the target and Newton-recovered aerofoil outlines, visually indistinguishable at full-chord scale, with the LE-region inset showing the same at the scale where FM-3 (§3) would be visible if the prescribed-LE mitigation were absent. It also includes a per-coefficient recovery-error bar chart making explicit which two of eighteen coefficients are pinned rather than recovered (§2.7's free-16/prescribed-2 framing). Figure 7 compares target and recovered C_p both as a full curve and restricted to the 16 QR-pivoted stations the Newton system actually solved against, confirming the coefficient-level agreement of Table 3 is not an artifact of comparing at only the fitted stations. Figure 8 shows the converged, released (natural-transition) flow solution at the recovered geometry, C_p and boundary-layer displacement thickness δ^*/c, the physical state the release-and-verify Δc_l check of Table 3 was computed from.

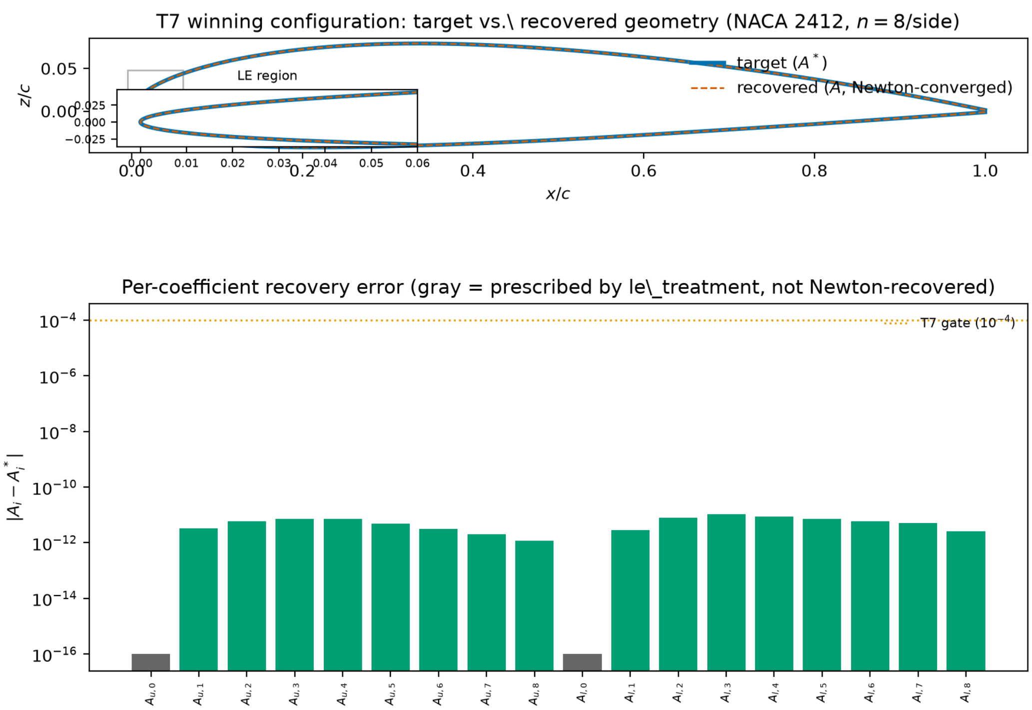

Figure 6: Target vs. Newton-recovered geometry, T7 winning configuration (NACA 2412, n=8/side), with LE-region inset and per-coefficient recovery-error bar chart (gray = prescribed by `le_treatment`, not Newton-recovered). Source: `experiments/results/t8/figures/paper/fig_t7_geometry_overlay.png`.

5 Ablations and results

Every number in this section traces to a `result.json` or `diagnostics.json` artifact under `experiments/results/t8/`. All values are N=1 (single `seed=42` run per cell); no confidence intervals are reported because none are statistically defensible at N=1, per [Sathish, 2026]'s Stage-1-lite scoping. The n-sweep, station-selection, and other single-factor *ablation* tables in this section are a

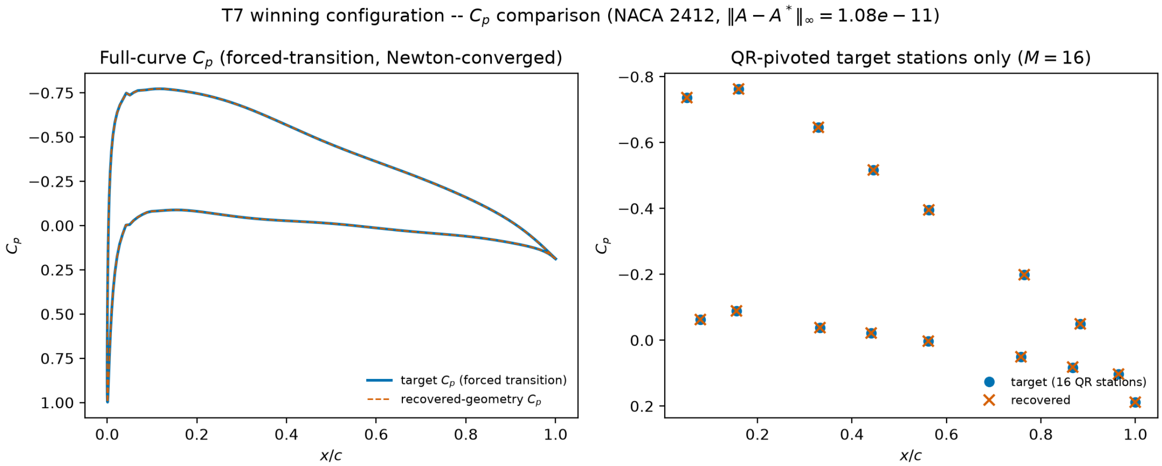

Figure 7: Target vs. recovered-geometry C_p (forced-transition, aero sign convention: y-axis inverted): full curve (left) and the 16 QR-pivoted target stations only (right). Source: `experiments/results/t8/figures/paper/fig_t7_cp_comparison.png`.

single-airfoil (NACA 2412) matrix plus two extra single-airfoil cells (0012, 4415). These are deliberately narrow, N=1-per-cell comparisons designed to isolate one factor at a time, not a panel study. Panel-level convergence-*fraction* language is deliberately avoided in those subsections for that reason. §5.5 is the one exception: it reports the pre-registered ≈20-section NACA panel (`configs/experiments/panel_naca/*.yaml`), run subsequently at the winning configuration identified by this section's ablations, and its Wilson-interval convergence-fraction claim is reported there on its own terms.

Station-addressing provenance. The ablation matrix in this section was run with target stations addressed by panel-node index, which was the implementation at the time. Addressing was subsequently changed to (surface, x/c) with interpolation between the two bracketing nodes, because a node index stops denoting the same physical location once the geometry moves, which is precisely what an inverse solve does (§2.3). The change moves the T7 figures quoted in §4 from 1.08×10^{-11} in 4 iterations to 2.75×10^{-11} in 6, both far inside the 10^{-4}/9-iteration gate, and leaves the QR-selected submap conditioning unchanged at 90.3. Every cell in this section shares the earlier addressing, so the within-matrix comparisons that the ablations rest on are unaffected; what must not be done is to read a single number from this section alongside a single number from §4 as though they came from one run. Where that distinction matters, it is stated at the point of use.

5.1 Station-selection identifiability: the central methodological result

This finding leads the ablation discussion, ahead of the flow-solve count, because it is the most novel, most precisely quantified, and most consequential result in T8. The square extended Newton system (§2.3) demands exactly M target-station equations. With evenly-spaced stations, between- station Cp information is discarded, and the resulting 16×16 station- sensitivity submap can be badly ill-conditioned. Table 4 reports the two configurations that isolate this mechanism.

Both cells' Newton loops report their own internal `converged: true`; the residual reaches the solver's floor in both cases. **Only `qr_pivot` recovers the correct design.** "Converged" and "correct" are different claims, invisible unless err_all_inf is checked against the true coefficients

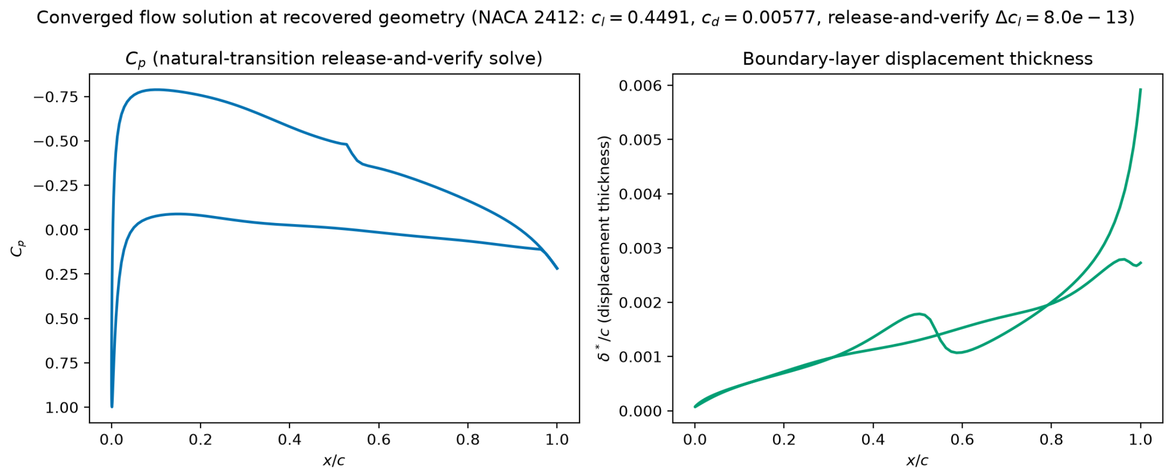

Figure 8: Converged flow solution at the recovered geometry: natural-transition release-and-verify C_p (left) and displacement thickness δ^*/c (right). Source: `experiments/results/t8/figures/paper/fig_t7_flow_solution.png`.

Table 4: Station-selection ablation, the paper's central methodological result. Both cells report `converged: true` and drive the residual to the solver floor, yet their recovery errors differ by eight orders of magnitude: only the QR-pivoted selection recovers the generating design, and the evenly-spaced selection converges to a spurious root while its station submap conditions at 3×10^{12}. Source: `experiments/results/t8/ANALYSIS.md` §4.

Selection	err_all_inf	Submap cond	Newton `converged` flag
`qr_pivot` (winning config)	$\mathbf{1.08 \times 10^{-11}}$	90.3	`true`
`even` (evenly-spaced)	$\mathbf{1.43 \times 10^{-3}}$ (FAILS 10^{-4} gate)	3.47×10^{12}	`true`

rather than the residual norm alone (experiments/results/t8/ANALYSIS.md §0, Deviation 5). The mechanism: with evenly-spaced stations, near-null coefficient directions exist whose motion changes the sampled Cp by less than floating-point precision. Newton converges to *any* of several exact roots of the (badly- conditioned) discrete system, a structural non-uniqueness of the sampled problem, not a numerical accident. This reproduces and sharpens a pre-T8 finding ([Sathish, 2026]: cond $\sim 10^7$, error 1.4×10^{-3}–6×10^{-2} with evenly-spaced stations) with a controlled ablation cell. Two contributing null-direction families were identified and eliminated *before* the station-selection fix, each verified empirically ([Sathish, 2026]): a camber–α equivalence (freeing α lets a slightly different camber at a slightly different α interpolate the same samples, so α is fixed for self-consistency tests), and a prescribed-LE mismatch (excluding LE stations while leaving the LE-dominant coefficients free leaves A_0 unobservable, which is why `le_treatment=prescribed` *fixes* $A_{u,0}/A_{l,0}$ rather than merely dropping LE-region target rows).

The resolution keeps the system square, with no least-squares relaxation and no reintroduced optimisation, and instead makes each of the M equations maximally informative. The T4 pre-solve (§2.7) already builds the full $(200 \times n_A)$ sensitivity matrix for initialisation; reusing it to *select* the M target stations by QR column pivoting chooses the rows that maximise the resulting square submap's conditioning. The measured effect: submap condition $10^7 \to 90.3$; recovery error $1.4 \times 10^{-3} \to 1.08 \times 10^{-11}$. Full-curve sensitivity information enters only through the (already-required) linear pre-solve, and the nonlinear Newton solve stays square and determined throughout ([Sathish, 2026]). Sensitivity-optimal station selection is therefore the identifiability guard for determined inverse systems, a methodological contribution of this paper beyond the original dossier hypothesis: a practical answer to the non-uniqueness caveat inverse problems inherit from Lighthill's and Volpe & Melnik's well-posedness constraints (§2.7; [design note Sathish, 2026, §7.10]), without abandoning the root-finding architecture for a least-squares relaxation.

Figure 9 summarises the ablation of Table 4: the same solver, the same target and the same convergence flag produce recovery errors eight orders of magnitude apart, separated only by which stations were sampled.

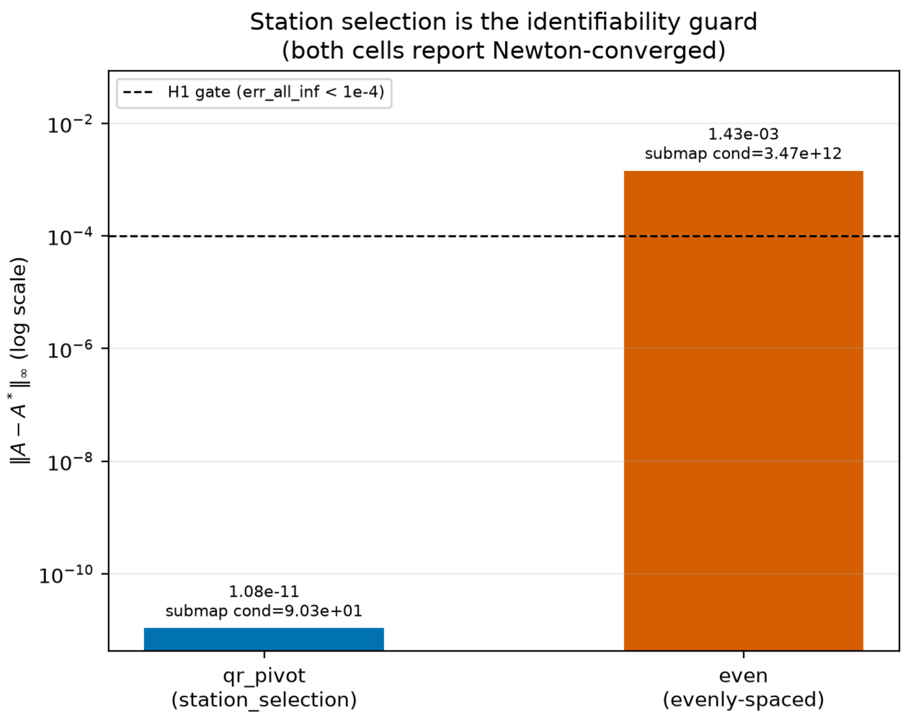

Figure 9: Station-selection identifiability at a glance: recovery error, QR-pivoted vs. evenly-spaced target stations, annotated with submap conditioning. Source: `experiments/results/t8/figures/station_selection.png`.

5.2 Where the stations are placed: the prescribed leading edge

§5.1 records one half of a requirement: excluding leading-edge target stations while leaving the LE-dominant coefficients free leaves A_0 unobservable, which is why `le_treatment=prescribed` fixes $A_{u,0}$ and $A_{l,0}$ rather than merely dropping LE-region rows. The other half is the converse, and it was found while extending the method to arbitrary user-supplied targets: once the leading edge *is* prescribed, the LE-region target stations must in turn be excluded. The two conditions are not independent choices. Prescribing the nose without restricting the candidate stations, or restricting the stations without prescribing the nose, each leaves a near-null direction in the sampled system; only the pair is identifiable.

Under the prescribed-LE treatment (FM-3, §3), $A_{u,0}$ and $A_{l,0}$ are given rather than solved, so a target row placed inside the prescribed region constrains pressure over a piece of surface that cannot move. The effect on sampling is not a uniform loss of information, and Figure 10(a) is worth reading carefully on this point. Across the prescribed region the norm of $\partial C_p/\partial A_{\text{free}}$ spans nearly two orders of magnitude, from 0.054 at the nose itself to 3.90 just aft of it, against a median of 1.47 over the rest of the chord. The region therefore contains both the single least informative station on the surface and some of the most sensitive. That combination is what makes unrestricted QR selection a poor choice here rather than merely a neutral one: the pivoting is drawn to the high-sensitivity nose stations, and the rows it selects there are strongly mutually correlated, because a small neighbourhood of a fixed nose produces nearly the same pressure response for every free coefficient. The extended system stays formally square and full-rank while becoming numerically near-dependent, which is the same underlying condition as Table 4 reached from the opposite direction.

Restricting the QR candidate set to $x \geq$ the prescribed-LE fraction, as Algorithm 2 specifies, removes the degeneracy. Table 5 reports the effect on a viscous self-consistency target, holding everything else fixed. The comparison is run at $n = 6$ per side, not the $n = 8$ of the ablation matrix above, so it is a controlled single-factor experiment in its own right rather than an additional cell of that matrix; its numbers should not be read against the $n = 8$ rows of Table 6.

Table 5: Target-station placement under a prescribed leading edge, at Bernstein order $n = 6$ per side rather than the $n = 8$ used elsewhere in this section. Both cells start from the same seeded perturbed geometry and differ only in whether the QR candidate set is restricted to $x \geq$ the prescribed-LE fraction. With the leading-edge region left in the candidate set the solve does not converge at all within the 50-iteration budget, despite driving its residual to 1.9×10^{-10}. Surface offset is quoted in millichords, against the 1 millichord that §2's fit gate allows a CST representation itself. Source: `experiments/results/le_stations/result.json`.

Quantity	LE stations in candidate set	LE stations excluded
Candidate stations	200	169
Submap condition	31.7	19.9
Converged	**no**	**yes**
Newton iterations	50 (budget exhausted)	**7**
Final residual	1.89×10^{-10}	8.21×10^{-11}
$\lVert A - A^* \rVert_\infty$ (free)	6.97×10^{-1}	$\mathbf{4.10 \times 10^{-3}}$
Max surface offset (mc)	11.74	**0.127**

The residual is not what separates the two columns. The unrestricted cell drives its residual to 1.9×10^{-10}, within an order of magnitude of the restricted cell's, and still fails to converge inside a 50-iteration budget while ending 11.7 millichords from the target surface. Restricting the candidate set converges in 7 iterations to 0.127 millichords, a ninety-fold improvement in the recovered shape, with the submap conditioning improving only from 31.7 to 19.9. A conditioning number that moves by less than a factor of two therefore does not bound the effect: what matters is whether the equations are spent on a part of the surface the solver is allowed to move.

This reinforces rather than repeats §5.1. There, the degeneracy came from discarding between-station information across the whole chord, and both configurations converged, one of them to a wrong root. Here it comes from spending equations on a region that has been removed from the design space, and the failure is visible as non-convergence rather than as a confident wrong answer. Neither is detectable from the residual norm alone.

5.3 Bernstein order n: the FM-2 conditioning cliff

Table 6 sweeps Bernstein order n (per side) from 4 to 12, reporting the three ADR-0004-distinct conditioning quantities alongside iteration count and recovery error. These are distinct matrices,

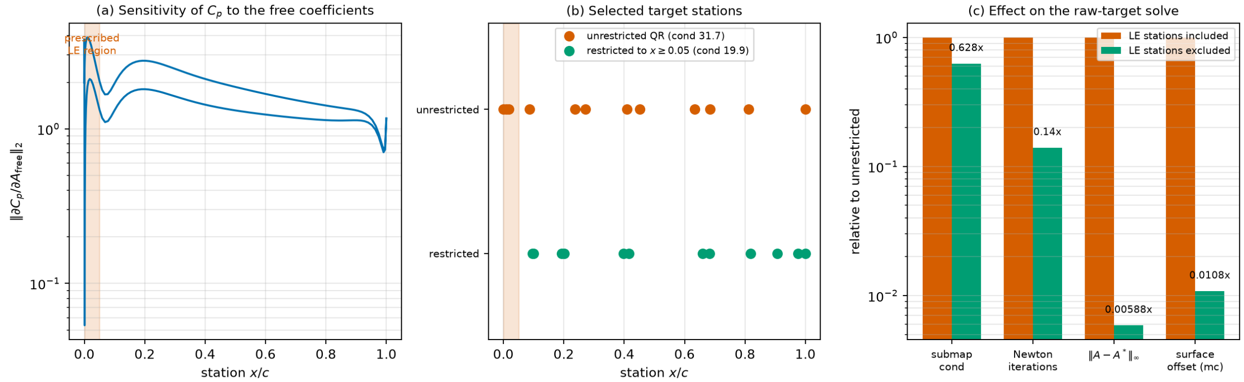

Figure 10: Why target stations must be kept out of a prescribed leading edge. (a) the sensitivity of surface pressure to the free coefficients across the chord, with the prescribed region shaded: it does not vanish there but varies by nearly two orders of magnitude, holding both the least and some of the most sensitive stations; (b) the stations chosen by unrestricted QR against those chosen from the restricted candidate set, with the resulting submap conditioning; (c) the effect on the solve, plotted relative to the unrestricted case on a logarithmic axis. Values as in Table 5.

plotted and reported separately per §2.7's reporting rule.

Table 6: Bernstein order sweep, $n = 4$ to 12 per side. Iteration count is flat at 3 to 4 through $n = 10$ and then jumps to 21 at $n = 12$, a cliff rather than a gradual degradation, while all three conditioning measures climb monotonically throughout. Source: `experiments/results/t8/ANALYSIS.md` §4.

n	Iterations	err_all_inf	Submap cond	cond(J), D-2, it. 0	T2 Gram/fit cond
4	3	8.08×10^{-12}	10.6	6.46×10^{7}	498
6	4	1.59×10^{-10}	20.4	8.46×10^{7}	7,327
8	4	1.08×10^{-11}	90.3	1.57×10^{8}	108,538
10	4	1.46×10^{-10}	592	4.52×10^{8}	1,626,188
12	**21** (FAILS 9-iter gate)	6.31×10^{-10}	1,144	2.07×10^{9}	24,579,174

Iteration count is flat at 3–4 through $n = 10$, then jumps to 21 at $n = 12$. This is a genuine cliff, not gradual degradation, though it tracks (without being identical to) all three independently-measured conditioning series, which climb monotonically with n throughout the sweep (roughly 4^n-like growth in the T2 Gram/fit condition, consistent with the a priori FM-2 hypothesis, [design note Sathish, 2026, §4] FM-2). $n = 12$ still eventually reaches the accuracy gate ($6.31 \times 10^{-10} < 10^{-4}$); it fails only the iteration-count budget, which is itself informative: FM-2 degrades convergence *speed* before it degrades final accuracy. Figure 11.

5.4 FM-1, FM-4, and other single-factor ablations

Table 7 reports the remaining single-factor ablations against the `t8_n08_baseline` cell.

The two DOF-mis-squared cells fail at construction time with an explicit, correct error string and zero Newton iterations spent. The FM-1 guard (Eq. (11)) catches deliberate over/under-determination cleanly: no silent corruption, no wrong answer (experiments/results/t8/ANALYSIS.md §4).

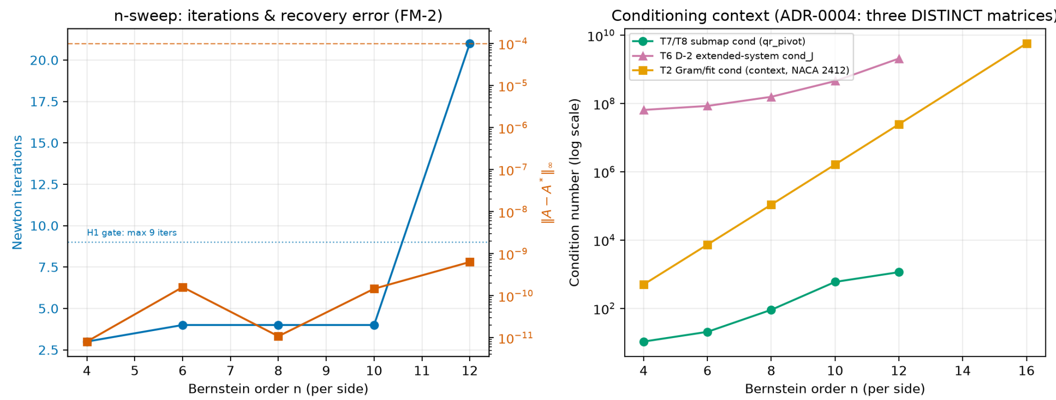

Figure 11: Iterations and err_all_inf vs. Bernstein order n, with the three ADR-0004-distinct conditioning series plotted separately. Source: `experiments/results/t8/figures/n_sweep.png`.

5.5 H1: NACA panel generalization

Table 8 adds two additional single-airfoil cells (0012, 4415) to the 2412 baseline reported throughout §4; that 3-airfoil table alone is N=1 per section and explicitly not a generalization claim. It was subsequently superseded by the pre-registered H1 evaluation itself, run post-review after the T8 ablation matrix closed (experiments/results/t8/ANALYSIS.md, "H1 addendum"): the winning configuration (§4) applied to a 20-section NACA panel spanning 9–25% thickness and 0–6% camber across both 4- and 5-digit families (`configs/experiments/panel_naca/*.yaml`, `seed=42`).

Result: 18/18 generable sections recovered, at err_free_inf $\leq 1.51 \times 10^{-10}$ **in 3–7 Newton iterations each.** Two of the 20 pre-registered sections produced no result and are excluded under the pre-registered §6-class exclusion rules (experiments/results/t8/ANALYSIS.md, "H1 addendum," Deviation 11), not counted as failures: For `panel_0006`, the forced-trip *target* generation itself fails to converge for this 6%-thin section at a 5% trip location, so no inverse problem was ever posed (a direct-solve non-convergence exclusion, exactly the pre-registered rule's intended case, not an architecture failure). For `panel_44012`, the vendor NACA 5-digit geometry generator does not implement the 44XXX mean-line family, a pre-registration defect (the target list included a section outside the implemented geometry generator), not a solver failure. Figure 12 shows 6 representative recovered sections spanning this range: thin symmetric (0009), thick symmetric (0025), the cambered baseline (2412), a thick cambered section (4415), a standard (non-reflexed) 5-digit-camber NACA5 (23012), and a high-camber section (6412). Target and recovered outlines are visually indistinguishable at this scale, consistent with the $\leq 1.51 \times 10^{-10}$ coefficient-level agreement.

Interval-arithmetic limits of the panel size (the pre-registered criterion still fails, on a technicality). At $n_{\text{eff}} = 18$ successes out of 18 eligible trials, the Wilson 95% lower bound is **0.824**, *below* the pre-registered criterion (Wilson LB ≥ 0.9), even though every eligible section succeeded. This is not a solver shortfall. A Wilson lower bound of 0.9 is mathematically unattainable at $n = 18$ or $n = 20$ regardless of outcome (the smallest all-success sample whose Wilson lower bound clears 0.9 is $n = 35$, at 0.901; $n = 29$ reaches only 0.883): the pre-registered target and the pre-registered panel size are mutually inconsistent, a pre-registration defect (experiments/results/t8/ANALYSIS.md, Deviation 12). Treating the excluded `panel_0006` as a failure rather than an exclusion instead gives 18/19 (0.947), Wilson LB 0.754, still short of 0.9, for the same

Table 7: Single-factor ablations vs. baseline. Source: paper/results_digest.md §5.2.

Factor	Cell	Iterations	err_all_inf	Notes
Transition free vs. forced	`t8_transition_free`	4	1.28×10^{-11}	no material difference (FM-4 mitigation cost is negligible once applied)
α free vs. fixed	`t8_alpha_free`	5	2.31×10^{-9}	1 extra iteration; consistent with the camber–α null direction (§5.1)
Init: perturbed vs. presolve	`t8_init_perturbed`	5	2.12×10^{-11}	43 vs. 120 flow-solve-equiv (§5.7); presolve buys efficiency, not correctness
LE treatment: none vs. prescribed	`t8_le_none`	4	6.50×10^{-10}	submap cond roughly doubles (195.6 vs. 90.3), still passes gate by 6 orders of magnitude
DOF over-determined (+1)	`t8_dof_over`	0 (clean fail)	n/a	Eq. (11) violated; `"M+K=17 but n_A_free+n_alpha=16"`
DOF under-determined (−1)	`t8_dof_under`	0 (clean fail)	n/a	Eq. (11) violated; `"M+K=15 but n_A_free+n_alpha=16"`

Table 8: Three-airfoil detail (baseline + 2 extras). Source: paper/results_digest.md §5.2.

Airfoil	Iterations	err_all_inf	Submap cond	Model gap
2412 (baseline)	4	1.08×10^{-11}	90.3	0.110
0012	4	3.15×10^{-11}	83.9	0.083
4415	3	4.35×10^{-11}	117.9	0.163

n-too-small reason. On the NACA panel alone, the architecture converges and recovers the true coefficients on every panel section it could be legitimately posed on, and the formal pre-registered statistical criterion fails on interval arithmetic, not on solver performance. §5.6 reports the pre-registered UIUC extension, sized to resolve the interval-arithmetic limit, and gives the combined H1 verdict across both panels.

5.6 H1: UIUC panel extension

Table 9 and Table 10 report the pre-registered UIUC panel (STATS_PROTOCOL §3.3). Sections were drawn from the 123-file UIUC corpus, stratified by thickness and camber decile, after a pre-filter removed 6 sections whose n=8 CST fit exceeds the T2 gate (`configs/experiments/panel_uiuc/*.yaml`), leaving 117 cells. Results are in `experiments/results/t8/uiuc_panel_summary.json`, regenerable with `python -m cins.benchmarks.panel_summary`.

Result: accuracy holds on every converged section; the composite criterion fails on iteration count. Of the 117 cells, 24 pose no inverse problem because the target solve itself does not converge, so no target pressure distribution exists to match. Of the remaining 93 attempted cells, 83 converge and 10 fail during the attempt. All 83 converged cells recover the free coefficients

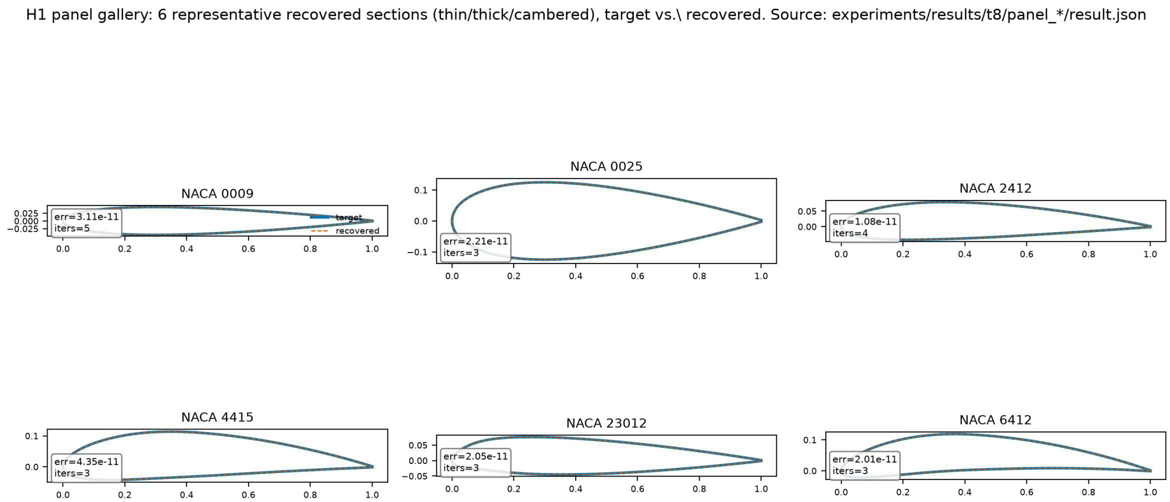

Figure 12: H1 panel gallery: 6 representative recovered sections (thin/thick/cambered/NACA5), target vs. recovered geometry, annotated with each cell's err_all_inf and iteration count from its `result.json`. Source: `experiments/results/t8/figures/paper/fig_h1_panel_gallery.png` (regenerated by `src/cins/benchmarks/paper_figures.py::fig_h1_panel_gallery`); scalar annotations from `experiments/results/t8/panel_*/result.json`.

to err_free_inf $< 10^{-4}$, a 1.000 success fraction with Wilson 95% lower bound 0.956. The iteration-budget criterion, ≤ 9 Newton iterations, holds for 73 of the 83 converged cells (0.880, Wilson LB 0.792). The pre-registered composite criterion, both accuracy and iteration budget together, is $73/83 = 0.880$ among converged cells (Wilson LB 0.792) and $73/93 = 0.785$ among attempted cells (Wilson LB 0.691), both below the pre-registered 0.9 threshold.

Ten converged sections exceed the iteration budget, requiring between 10 and 50 Newton iterations while still recovering coefficients to between 1×10^{-11} and 3×10^{-10}, the same accuracy band as the sections that converge within budget. The single-digit iteration expectation in the dossier was calibrated on NACA sections (§5.5). It does not transfer to the wider UIUC geometry space, where sections with unusual camber or thickness distributions converge more slowly without loss of final accuracy.

Exclusion classes. Table 11 lists the exclusion classes spanning the 24 not-posed and 10 attempt-failure cells.

The `target_natural` and `target_forced` classes account for 23 of the 24 not-posed cells: the flow solver cannot produce a converged solution for the section itself, so no target exists and the inverse formulation is not exercised. The `geometry_te_gap` class covers three sections, `goe780`, `mh62`, and `naca64206`, that fail mfoil's wake-direction assertion (`vendor/mfoil/mfoil.py`, `build_wake`) after the minimum-trailing-edge-gap fix that resolved the sharp-trailing-edge case. Their trailing-edge gap is already above the minimum, so the cause is distinct from the case the fix addressed. This is recorded as an open item in §6.2 rather than resolved here.

Table 9: UIUC panel outcome by population. Source: experiments/results/t8/uiuc_panel_summary.json; experiments/results/t8/ANALYSIS.md, "H1 addendum, part 2."

Population	Definition	Count
Cells	Sections with a generated configuration	117
No inverse problem posed	Target solve did not converge, so no target pressure distribution exists	24
Attempted	Cells where a target existed and the inverse was run	93
Converged	Newton reached the residual tolerance	83
Attempt failures	Solver or geometry faults during the attempt	10

Table 10: UIUC panel criteria and Wilson 95% lower bounds. Source: experiments/results/t8/uiuc_panel_summary.json.

Criterion	Result	Wilson 95% lower bound
Coefficient accuracy (err_free_inf $< 10^{-4}$), among converged	$83/83 = 1.000$	0.956
Iteration budget (≤ 9 iterations), among converged	$73/83 = 0.880$	0.792
Both criteria, among converged	$73/83 = 0.880$	0.792
Both criteria, among attempted	$73/93 = 0.785$	0.691

Combined H1 verdict (NACA + UIUC). Accuracy holds on every section that converged across both panels: 18 of 18 eligible NACA sections and 83 of 83 converged UIUC sections recover the free coefficients to err_free_inf $< 10^{-4}$, with observed errors clustering between 10^{-11} and 10^{-9}. The pre-registered composite criterion, which also required the solve to complete within 9 Newton iterations, is not met on the UIUC panel: the Wilson lower bound on the composite criterion is 0.792 among converged UIUC cells and 0.691 among attempted UIUC cells, both below the pre-registered 0.9 threshold. The cause is the iteration budget, not accuracy. The single-digit iteration expectation in the dossier was calibrated on NACA sections and does not transfer to the wider UIUC geometry space, where sections with unusual camber or thickness distributions take longer to converge while still reaching the same accuracy. **H1 verdict:** the hypothesis that the monolithic formulation recovers geometry on realisable targets is supported on accuracy across both panels, 18/18 NACA sections and 83/83 converged UIUC sections out of a 117-section UIUC panel. The pre-registered composite criterion, which also required single-digit iterations, is not met on the UIUC panel. The failure is isolated to the iteration-count component; the accuracy component holds on every converged section in both panels.

5.7 Flow-solve count vs. nested optimisation

The corrected headline. STATS_PROTOCOL pre-registered a ≥100x flow-solve-reduction claim; it is **rejected** under both fair-paired controls actually run. Table 12 pairs each monolithic run against a nested Levenberg–Marquardt control sharing the same initialisation strategy, so the comparison is apples-to-apples in the same currency (`n_flow_solves_equivalent`, both sides).

The nested control is `scipy.optimise.least_squares(method="lm", ftol=xtol=gtol=1e-8, x_scale="custom"(`$|x_0|$` floored at `10^{-3}`))` [Virtanen et al., 2020], converged both times (`xtol` termination), over the same 16 free coefficients. This is a competent modern LM solver, not a weak strawman: it converges the cold-start case in `n_nfev=6` outer LM iterations, each costing multiple flow solves for its own finite-difference Jacobian columns. **Recommended P1 sentence**: *"The monolithic architecture requires 3.1–8.1x fewer flow-residual evaluations than a fairly-paired, well-*

Table 11: UIUC panel exclusion classes. Source: experiments/results/t8/uiuc_panel_summary.json.

Exclusion class	Count
`target_natural`	16
`target_forced`	7
`solver_stagnation`	3
`other`	3
`geometry_te_gap`	3
`other:RuntimeError`	1
`init_flow_solve`	1

Table 12: Flow-solve-count comparison, monolithic vs. nested LM control. Source: paper/results_digest.md §5.3; experiments/results/t8/ANALYSIS.md §2.

Pairing (shared init strategy)	Monolithic solves	Nested LM solves	Ratio
Presolve-init	120	373	**3.1**×
Perturbed-init (cold-start)	43	347	**8.1**×

tuned nested Levenberg–Marquardt baseline (`scipy.least_squares`), depending on initialisation strategy, substantially less than the order-of-magnitude reduction hypothesised a priori, though the qualitative advantages of determinism, guaranteed local quadratic convergence, and exact analytic constraint-row Jacobians persist independent of this ratio." Figure 13.

5.8 Presolve vs. station selection as the uniqueness guard

`t8_init_perturbed` (no presolve, `station_selection= qr_pivot`): 43 flow-solve-equivalents, 5 iterations, error 2.12×10^{-11}, submap cond 89.8, essentially identical to the baseline's 90.3, since station selection is a property of the stations (computed once from the T4 sensitivity matrix), not of the initial guess. `t8_n08_baseline` (presolve, same station selection): 120 flow-solve-equivalents, 4 iterations, error 1.08×10^{-11}, submap cond 90.3. Interpretation: skipping the presolve costs one extra Newton iteration but *reduces* total flow-solve count ($43 < 120$) and does not cost correctness. The presolve consumes roughly 77 flow-solve- equivalents (≈64% of the baseline total), primarily for iteration efficiency and starting-residual quality, not for the uniqueness guarantee, which comes from `station_selection=qr_pivot` alone. This has *not* been verified in the sharper crossed form (`init=perturbed` × `station_selection=even`); it is flagged explicitly as future work, not as tested (experiments/results/t8/ANALYSIS.md §5, "Caveat").

5.9 Quadratic convergence tail

STATS_PROTOCOL's 3-point local order estimator ($p \approx \log(R_k/R_{k-1})/\log(R_{k-1}/R_{k-2})$) is applied only where ≥ 3 residual values exist strictly above the $\approx 10^{-9}$ FD-noise floor. The raw `convergence_order` field stored per cell does *not* apply this exclusion and is not quoted directly. Table 13 reports the recomputed orders for the five cells with enough clean-tail residuals to estimate one.

Median order: 2.15 (n=5 estimable cells), meeting the pre- registered ≥ 1.8 target. The remaining 7/12 converged cells cannot be scored by this method because they reach the solver's

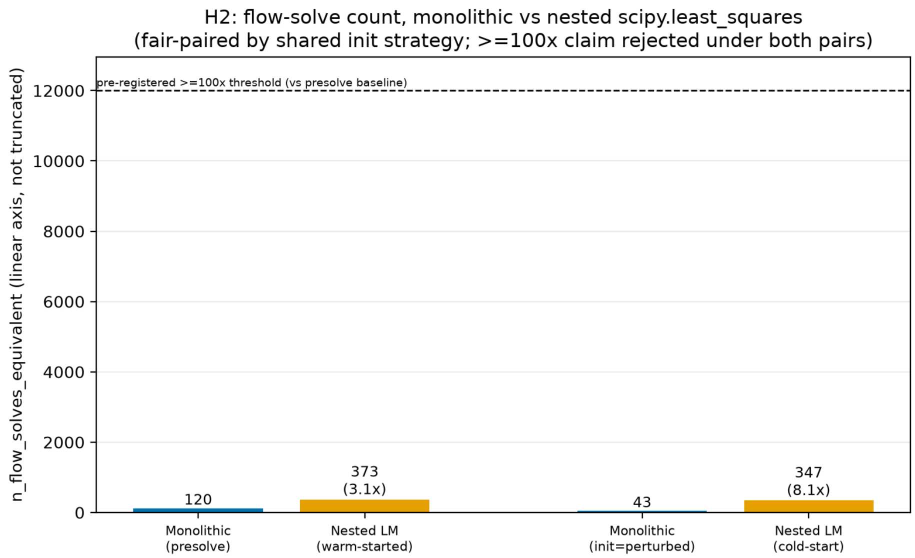

Figure 13: Flow-solve-count comparison, linear non-truncated axis, both fair-paired comparisons, pre-registered 100x line shown for scale. Source: `experiments/results/t8/figures/h2_flow_solves.png`.

Table 13: Recomputed convergence order, clean (floor-excluded) tail. Source: paper/results_digest.md §5.5; experiments/results/t8/ANALYSIS.md §3.

Cell	Recomputed order (clean tail)
`t8_alpha_free`	1.09
`t8_init_perturbed`	2.08
`t8_n06`	2.46
`t8_transition_free`	2.15
`t8_n12`	4.24 (late-tail 3-point estimate; read as "clearly superlinear," not literal quartic order)

residual floor in only 2 post-initial steps, too fast for a 3-point estimator to have 3 clean points. This is itself consistent with (though it does not formally confirm) fast superlinear-to-quadratic convergence: the baseline, for example, drops from $O(10^{-4})$ to the $\sim 10^{-10}$ floor in exactly 2 Newton steps, a $\sim$150x-per-step reduction rate not consistent with linear convergence (experiments/results/t8/ANALYSIS.md §3).

5.10 Comparative positioning against other inverse-design approaches

Tables 14 and 15 place this paper's architecture against five other approaches to the same underlying problem (given a target surface pressure or speed distribution, recover a geometry that produces it), each representing a structurally distinct formulation class. The comparison is split across two tables because its two halves are not commensurable in the same way: Table 14 collects the quantitative axes, where the cost column is reported in whatever unit each formulation class naturally exposes, and Table 15 collects the structural guarantees, where the classes separate sharply and no numbers are involved at all. Rows (i) and (ii) are this paper's own measured, fair-paired numbers (§5.7); rows (iii) to (vi) are drawn from the cited literature and are explicitly marked *qualitative* wherever the source does not report a directly comparable metric. No number in either table is asserted unless it is a repository artifact from rows (i) or (ii), or attributed to a specific cited source.

Three patterns follow from the table. First, only rows (i) and (iii) are architecturally root-finding rather than optimisation or amortised inference. This distinction, argued in this paper's Introduction (§1.1), is not cosmetic: a converged root-find has a residual-norm certificate that a converged optimisation or a sampled generative output does not. Second, the flow-solve-count column spans roughly four orders of magnitude across the table, not because the methods disagree about a shared quantity, but because "flow solve" means a different thing in each formulation class (a Newton linearisation vs. an LM outer iteration vs. an offline training epoch vs. a primal/adjoint pair). This is exactly the currency-matching discipline §5.7 enforces for the one comparison (row (i) vs. row (ii)) this paper actually measured; rows (iii)–(vi) are reported qualitatively for this reason, not omitted. Third, the licence/availability column is not incidental context: it is the practical reason MISES's built-in Modal-Inverse mode (row (iii)), despite being architecturally identical to this paper's contribution, has not seen the open comparative study this table attempts. A $10k-gated, non-open-source solver is not reproducible science. This is also why mfoil [Fidkowski, 2022] rather than MISES is this paper's Stage 1 testbed (§2), and why the Stage 3 extension (§7) targets porting the architecture *into* MISES rather than requiring readers to already have a MISES licence to reproduce this paper's own results.

6 Discussion

6.1 A common separation-principle pattern across the four mitigations

The FM-2/FM-3/FM-4 mitigations used in this paper are not independent engineering fixes. They are recognisable instances of a single TRIZ contradiction-resolution pattern, *separation*, applied along whichever axis resolves the specific improving/worsening parameter pair in tension ([Sathish, 2026]; [Sathish, 2026]):

- **FM-4 (forced trip, then release) is separation in time** (§2.6). The contradiction is reliability (a smooth, Newton-tractable residual) vs. model accuracy (the true e^n closure). Resolving it by phase, forced/smooth *during* inverse iterations, natural/accurate *after* convergence, with a feedback verification step, sacrifices no accuracy in the end while removing the C^0 kink exactly when Newton needs it removed.

- **FM-3 (prescribed-LE treatment) is separation in space**. The contradiction is design freedom (invert everywhere) vs. Jacobian-row conditioning (rows near the nose blow up). Prescribing the first ≈5% chord geometrically and inverting only aft of it removes the pathological rows entirely, at the cost of design freedom the manufacturing constraints (mechanical, cooling) of a real turbine blade were going to fix anyway.

- **FM-2 (orthogonalised basis over Tikhonov regularisation, and in practice, keeping n inside the pre-cliff range identified in §5.1) is TRIZ principle 3, local quality**. The tension identified in the dossier is that the natural fix for ill-conditioning, Tikhonov regularisation, turns a determined solve back into a least-squares problem, i.e. back into the optimisation this architecture exists to avoid. The discipline instead is to keep the basis order in its well-conditioned regime (or orthogonalise it) rather than regularise a poorly-conditioned one.

- The station-selection identifiability guard of §5.1 is likewise principle-3/16 in character (local quality, partial/excessive action): rather than adding more target equations (which would

Table 14: Comparative positioning, quantitative axes: formulation class, cost per design, and the precision each approach has been demonstrated to. Cost is reported in the unit each formulation class naturally exposes, which is why the column is not a single commensurable number; "qual." marks an entry for which no directly comparable published metric exists.

Method	Formulation class	Flow solves per design	Demonstrated precision
(i) **This work** (CINS monolithic, §2)	constrained root-finding, single square Newton system	**43–120** measured, T7 winning configuration, two initialisation strategies (§5.7)	$\|A - A^*\|_\infty$ at the 10^{-11} level on self-consistency (Table 3)
(ii) Nested LM baseline (`scipy.least_squares`, §5.7)	outer-loop nonlinear least-squares optimisation	**347–373** measured on the same paired cases (§5.7)	converges to `xtol` against the same coefficient-level target, paired with row (i)
(iii) MISES Modal-Inverse [Drela, 1990]	constrained root-finding, mode-shape amplitudes appended to the same global Newton system	one Newton solve per design, architecturally identical to row (i); exact count not published	qual.: production-validated on transonic cascades over decades, no single benchmark number available for direct comparison
(iv) CST-smoothing inverse [Lane and Marshall, 2010]	iterative outer loop; CST used as a *post-hoc smoother* on a pressure-residual-driven shape correction, solver-agnostic	qual.: multiple outer iterations, count depends on the wrapped solver and is not reported as a fixed number in the source	qual.: subsonic and transonic cases shown converging; no coefficient-recovery benchmark reported
(v) Learned generative inverse (cGAN [Yilmaz and German, 2020]; CDDPM [Yu et al., 2025]; LF-PINN [Wang et al., 2024])	amortised statistical inference, offline-trained generative or surrogate map	approximately zero *online* per design after training; training-set generation requires $O(10^3\text{–}10^5)$ offline flow solves, typical for this literature	CDDPM reports a 34.4 percent precision improvement over WGAN baselines [Yu et al., 2025], a relative rather than absolute geometric-recovery metric; LF-PINN reports at least a tenfold online speedup against direct mfoil analysis [Wang et al., 2024]
(vi) Adjoint target-C_p optimisation (SU2 [Economon et al., 2016])	gradient-based PDE-constrained optimisation, primal and adjoint pairs per design-cycle iteration	qual.: $O(10\text{–}100)$ primal/adjoint pairs typical for shape optimisation of this scale, case- and tolerance-dependent	qual.: mature and widely validated on shape-optimisation benchmarks; not run in this study, so no head-to-head number is available

Table 15: Comparative positioning, structural axes: what each approach guarantees about the solution it returns, how it handles geometric design constraints, and its availability. The determinism column is the one on which the formulation classes separate most sharply.

Method	Determinism and uniqueness guard	Constraint handling	Licence and availability
(i) **This work** (CINS monolithic)	deterministic Newton; QR-pivoted station selection is an explicit, measured identifiability guard (§5.1)	exact linear equality rows $GA = b$, with a zero-cost constant Jacobian block	open [Sathish, 2026]; mfoil is MIT-licenced [Fidkowski, 2022]
(ii) Nested LM baseline	deterministic given fixed initialisation and tolerances; no analogous uniqueness guard, relying on local LM convergence only	none built in; bounds and scaling only, with no analytic constraint rows in this configuration	open [Virtanen et al., 2020]
(iii) MISES Modal-Inverse	deterministic Newton; no published uniqueness or identifiability study of mode count against target sampling analogous to §5.1	mode amplitudes are heuristic bump functions rather than analytic geometric functionals (§1.1)	gated: free for faculty-signed academic use, commercial licence otherwise [Drela, 1990]; no open-source port exists
(iv) CST-smoothing inverse	deterministic given fixed settings; no formal uniqueness guard reported	no analytic constraint rows; the correction is smoothed but not constrained to a design-parameter manifold	method description only, solver-agnostic, with no released code cited
(v) Learned generative inverse	*not* deterministic for GAN and diffusion variants, which sample stochastically; LF-PINN's surrogate is deterministic but is an approximation rather than an exact solve. **No convergence guarantee or exact-recovery certificate exists for any variant.**	none intrinsic to generation; plausible-geometry constraints are learned from the training distribution rather than enforced algebraically	open-source code released with most cited papers, but a training database is required
(vi) Adjoint target-C_p optimisation	deterministic gradient descent to a *local* optimum; no uniqueness guard, which is the sense in which adjoint target-C_p matching remains optimisation rather than root-finding [design note Sathish, 2026, §6.3]	general nonlinear constraints via adjoint sensitivities: flexible, but iterative rather than closed-form	open source, LGPL [Economon et al., 2016]

over-determine the square system) or accepting whichever stations are convenient, the existing T4 sensitivity matrix is reused (principle 25, self-service) to select which M rows are maximally informative.

Each mitigation was arrived at independently, from direct numerical evidence: a stalled residual, a spiked row-norm, a badly conditioned submap. The common resolution structure was recognised only afterward.

6.2 Limitations

- **Single-airfoil-family, N=1 *ablations* (the n-sweep, station-selection, and other single-factor tables of §5.1–§5.7).** These remain NACA-2412-only, `seed=42`, one run per cell: Stage-1-lite evidence by design, with no confidence interval reported for any of them. This caveat does *not* extend to the H1 convergence claim itself (§5.5, §5.6), which was subsequently upgraded to genuine panel-level evidence across both the 20-section NACA panel (18/18 generable sections, Wilson LB 0.824) and the 117-section UIUC panel (83/83 converged sections meet the accuracy criterion, Wilson LB 0.956). The ablation-level findings (n-sweep, station-selection, flow-solve-count) remain NACA-2412-only and are not extended to the UIUC panel by this paper.
- **Vendor wake-direction assertion failures beyond sharp trailing edges.** Three UIUC sections, `goe780`, `mh62`, and `naca64206`, still fail mfoil's wake-direction assertion (§5.6) after the minimum-trailing-edge-gap fix that resolved the sharp-trailing-edge case. Their trailing-edge gap is already above the minimum, so the cause is distinct from the case the fix addressed. This is recorded as an open item, not resolved here.
- **Subcritical, isolated airfoil.** All results use mfoil's Kármán–Tsien compressibility correction (subcritical only) on an isolated airfoil: no cascade periodicity, no pitch/stagger, no AVDR streamtube contraction ([design note Sathish, 2026, §6.2]). The hypothesis under test is a claim about solver *architecture*, not turbine aerodynamics, and is answered as well on an isolated airfoil as on a cascade. The cascade and transonic regimes, however, are genuinely untested here.
- **Single operating condition.** All cells share one `operating.*` block (one Re, one α, one Mach regime); no claim in this paper generalizes across flow regimes.

6.3 The outdated-baseline finding

The pre-registered ≥100x claim (STATS_PROTOCOL H2) is not simply missed by the measured 3.1–8.1x. The two independent fair-paired controls in §5.7 give a plausible account of *why* it lands where it does. `scipy.optimize.least_squares(method="lm")`, tightly tolerance (ftol = xtol = gtol = 10^{-8}) and properly scaled (x_scale from $|x_0|$, floored at 10^{-3}) over the same 16 well-conditioned free coefficients, is a genuinely strong comparator. It converges the cold-start case in 6 outer LM iterations, nothing like the Dakota-surrogate/Kriging-in-the-loop nested-optimisation baselines that a ≥100x figure of that magnitude typically describes in the older aerodynamic-shape-optimisation literature the dossier's original framing appears to draw on. **The pre-registered threshold plausibly encodes an outdated class of nested-optimisation baseline.** Measuring against a strong modern gradient-based baseline and reporting the resulting 3–8x ratio directly is a more informative comparison than either confirming or missing the original number.

7 Conclusions

Appending CST coefficients as unknowns to a coupled viscous/inviscid Newton solver, with geometric design constraints expressed as the linear rows CST's own linearity makes available, converts constrained inverse airfoil design from a search over an objective function into a single square, determined root-find. The falsifiable self-consistency test (§4) passes with a wide margin on every criterion, including a release-and-verify check the original protocol had not yet required at the time of first closure. The T8 ablation matrix (§5) confirms the architecture's basic viability across three NACA sections and Bernstein orders up to a conditioning cliff at $n = 12$. It also *corrects* the a priori flow-solve-count hypothesis downward to a measured 3.1–8.1x, while surfacing a methodological contribution the original dossier did not anticipate: QR-pivoted station selection as the identifiability guard for square inverse systems. The paper's two central results are therefore a falsified numeric hypothesis (the flow-solve ratio) and a strengthened, unanticipated qualitative one (station-selection identifiability).

Future work. Three extensions are pre-planned ([design note Sathish, 2026, §8–§9], publication plan §10), none yet built. **Stage 2**, cascade periodicity, substitutes the panel-influence kernel $\ln(z - z_0) \to \ln[\sin(\pi(z - z_0)/s)]$ for pitch s (reducing to the isolated-airfoil kernel as $s \to \infty$, a free regression test), validated against Gostelow's exact cascade solutions before any turbine-cascade inverse-design claim is made. **Stage 3** carries the architecture into MISES itself via the Modal-Inverse `modes.xxx` mechanism. Because CST's perturbation modes are identical to its own basis functions (a direct consequence of the linearity in §2.1), this requires writing $C(\psi)S_i(\psi)$ into MISES's mode-file format, with no Fortran modification, and would demonstrate the hypothesis in a transonic, viscous, real-cascade solver at near-zero implementation cost if Route A succeeds. The pre-registered NACA/UIUC panel evaluation (STATS_PROTOCOL §3) is now complete (§5.5, §5.6): both panels support the accuracy component of H1, and the UIUC panel's composite criterion fails on the iteration-count component, not accuracy. The remaining gap between what this paper can currently claim and what was originally pre-registered is twofold: the Stage-1-lite, N=1 *ablation* evidence (§5.7's flow-solve-count comparison, the n-sweep, station-selection ablations) has not been extended to panel level. Second, three UIUC sections (`goe780`, `mh62`, `naca64206`) still fail the vendor wake-direction assertion for a cause unrelated to sharp trailing edges (§5.6, §6.2).

Data and code availability

All source, configurations, experiment manifests and figure-generation scripts are available at `https://github.com/SharathSPhD/CINS`. Every figure in this paper is regenerable from an archived run manifest recording the configuration hash, git SHA and random seed; the procedure is stated in Appendix D. The flow solver is `mfoil`, used unmodified as a vendored dependency.

Acknowledgements

The author thanks Krzysztof Fidkowski for releasing `mfoil` under the MIT licence, which made a Newton-coupled viscous solver with an accessible global system available for this work.

A Beta-function area-row derivation

§2.2 states the inscribed-area constraint row as a closed-form Beta function without deriving it; this appendix supplies the algebra (source: [design note Sathish, 2026, §3.4]). The per-surface area under Eq. (4) is

$$\mathcal{A}_{\text{surf}} = \int_0^1 \zeta(\psi)\,d\psi = \int_0^1 \Big[C(\psi)\sum_i A_i S_i(\psi) + \psi\,\zeta_T\Big]\,d\psi = \sum_i A_i \int_0^1 C(\psi)S_i(\psi)\,d\psi + \frac{\zeta_T}{2}, \quad (13)$$

using linearity of the integral (itself a consequence of Eq. (4)'s linearity in A, §2.1) and $\int_0^1 \psi\,d\psi = 1/2$ for the trailing-edge term. The remaining integral, per basis function $S_i(\psi) = K_i\psi^i(1-\psi)^{n-i}$ with $K_i = \binom{n}{i}$ and $C(\psi) = \psi^{N_1}(1-\psi)^{N_2}$, is

$$\int_0^1 C(\psi)S_i(\psi)\,d\psi = K_i \int_0^1 \psi^{N_1}(1-\psi)^{N_2}\,\psi^i(1-\psi)^{n-i}\,d\psi \quad (14)$$

$$= K_i \int_0^1 \psi^{\,i+N_1}(1-\psi)^{\,n-i+N_2}\,d\psi. \quad (15)$$

The Euler integral of the first kind is, by definition, $B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt$ for $\Re(p), \Re(q) > 0$. Matching exponents, $p-1 = i+N_1 \Rightarrow p = i+N_1+1$ and $q-1 = n-i+N_2 \Rightarrow q = n-i+N_2+1$, giving exactly Eq. (7):

$$\int_0^1 C(\psi)S_i(\psi)\,d\psi = K_i\,B(i+N_1+1,\; n-i+N_2+1). \quad (16)$$

For the default round-nose/sharp-tail class ($N_1 = 0.5$, $N_2 = 1.0$), this is $K_i\,B(i+1.5,\, n-i+2)$, always well-defined since $i+1.5 > 0$ and $n-i+2 > 0$ for every $i \in \{0,\dots,n\}$, so no term in the sum is singular despite the class function's own $\sqrt{\psi}$ non-analyticity at $\psi = 0$ (§2.2's point that the LE singularity affects flow- residual *rows*, not CST's own geometric *functionals*, which stay perfectly regular). Stacking one such row per coefficient gives the full area constraint row g in $g \cdot A = b$, with b the target inscribed area (or, for the T7/T8 self-consistency protocol, the area of the CST fit itself, per the target-consistent-RHS discipline $b = g \cdot A^*$ established at T4: the constraint right-hand side must always be built from what the reference geometry *actually has*, never from an idealised or rounded value, or the constraint row itself introduces a spurious residual floor no Newton iteration can remove).

B DOF accounting worked example (T7 configuration)

Eq. (11) ($M + K = n_{A,\text{free}} + n_\alpha$) is an assertion, not a derived quantity (§2.3); this appendix works it for the T7 winning configuration and states the general rule it specialises.

The T7 case. $n = 8$ per side $\Rightarrow$ 9 coefficients per surface, 18 total ($A_{u,0..8}$, $A_{l,0..8}$). `le_treatment=prescribed` removes 2 of these from the free set: $A_{u,0}$ and $A_{l,0}$ are geometrically fixed to the reference fit's own values, not Newton unknowns (§4), leaving $n_{A,\text{free}} = 16$. Because $A_{u,0}/A_{l,0}$ are prescribed directly rather than tied by an equality row, no separate shared-LE constraint row is needed for this configuration: $K = 0$. α is fixed for the self-consistency protocol (removing the camber–α null direction, §5.1): $n_\alpha = 0$. Eq. (11) therefore requires $M = 16 + 0 - 0 = 16$ target-station equations, exactly the QR-pivoted station count reported throughout §4–§5.1 (`experiments/results/t7_naca2412/diagnostics.json`, `static. dof_accounting`: `n_A_free=16, M=16, K=0`).

The general rule. Restating Eq. (11) in full generality: for n Bernstein coefficients per surface (2 surfaces, so $2(n+1)$ raw coefficients), $c_{LE} \in \{0,1\}$ constraint rows tying the leading-edge coefficients (shared-LE row, used only when `le_treatment=none` or `lem`, not when `prescribed` pins the coefficients directly), $p_{LE} \in \{0,2\}$ coefficients removed by prescription (2 when `le_treatment=prescribed`, else 0), and any additional geometric constraint rows K_{extra} (TE thickness, inscribed area, curvature, §2.2):

$$n_{A,\text{free}} = 2(n+1) - p_{LE}, \qquad K = c_{LE} + K_{\text{extra}}, \qquad M = n_{A,\text{free}} + n_\alpha - K. \tag{17}$$

This is the Stage-1 (isolated-airfoil) specialisation of the dossier's original rigid-body bookkeeping (§2.3's DOF $= 2(n+1) - 1_{\text{shared LE}} - 1_{\text{TE fixed}} + 3_{\text{scale/rotate/translate}} + 1_{\text{stagger or }\alpha}$): the three rigid overall modes are not exposed as free Newton unknowns in this implementation (chord, position and rotation are fixed by convention, not solved for), so they contribute neither to $n_{A,\text{free}}$ nor to M, and Eq. (11) is the resulting simpler, directly-assertable statement. The rule's lineage is Lighthill's own well-posedness bookkeeping [Lighthill, 1945]: a conformal-mapping inverse solve is well-posed only when the number of free shape parameters exactly matches the number of independent closure conditions (contour closure, LE radius, circulation) the target and geometry family jointly admit. Eq. (11) is that same counting argument, discretised for a finite-dimensional CST basis and asserted programmatically rather than checked by hand.

C The onset-closure Jacobian rows

§2.6 describes `_residual_transition_forced` (ADR-0003) narratively; this appendix states its Jacobian-row structure explicitly, since it is the one residual block in this paper's extended system that is not simply reused unchanged from mfoil (§2.3). At a forced-transition onset station with endpoint states U_1 (upstream, still formally laminar-indexed) and U_2 (downstream, freshly turbulent), the vendor three-row station residual is $R = [R_{\text{mom}}, R_{\text{shape}}, R_{\text{lag}}]$ (`mfoil.py`, `residual_station`). Rows 0–1 (momentum, shape parameter) are vendor's own ordinary turbulent closure evaluated directly across $[x_1, x_2]$: an unchanged code path, so their U- and x-Jacobian blocks are exactly mfoil's own $\partial R_{\text{mom,shape}}/\partial U_{1,2}$, $\partial R_{\text{mom,shape}}/\partial x$, reused verbatim. Row 2 (shear-lag) is replaced with the onset condition $U_2[2] - c_{ttr}(U_2) = 0$, where c_{ttr} is the same transition- correlation root-shear-stress function (`get_cttr`) vendor's own `update_transition` uses to initialise a newly-turbulent node. Because c_{ttr} is a function of U_2 (specifically $\theta, H_k, Re_\theta, u_e$ at the downstream node) only, its Jacobian row has the structure

$$R_U[2,\, 0{:}4] = 0 \quad \text{(no } U_1 \text{ dependence)}, \qquad R_U[2,\, 4{:}8] = e_{sa} - \left.\frac{\partial c_{ttr}}{\partial U_2}\right|_{U_2}, \qquad R_x[2,\, :] = 0, \tag{18}$$

where $e_{sa} = [0, 0, 1, 0]$ selects the shear-lag component of U_2 and $\partial c_{ttr}/\partial U_2$ is `get_cttr`'s own analytic derivative block (`cttr_U`), already computed by vendor code for the identical purpose inside `update_transition`: reused, not re-derived. The row's independence from U_1 and from x is not an approximation: it follows structurally from c_{ttr}'s own functional form, which was verified by direct inspection of `get_cttr` (ADR-0003, "Modeling choice"), and it is what makes this replacement row well-defined at a station where the natural (unforced) closure's internal x_t-solve has no solution (§2.6).

D Reproducibility statement

- **Repository.** https://github.com/SharathSPhD/CINS.git (branch `main` at every gate closure, per this project's workflow discipline). `vendor/mfoil/mfoil.py` [Fidkowski, 2022] is vendored unmodified (MIT licence) and is never edited; all solver access is through `src/cins/solver/mfoil_adapter.py`.

- **Manifest scheme.** Every experiment run in this paper writes an immutable manifest under `experiments/results/` recording the git SHA, a hash of the fully-resolved `configs/*.yaml` used, the random seed (`experiment.seed`, `42` throughout), and enough metadata to re-generate the run byte-for-byte ([Sathish, 2026] condition 4). No number in this paper is hand-edited; every table cell traces to a `result.json` or `diagnostics.json` artifact under `experiments/results/`, cited inline by path throughout §4– §5.

- **One-command regeneration.** T8 ablation-matrix figures (D-6 overlay, n-sweep, flow-solve bar chart, station-selection comparison): `.venv/bin/python -m cins.benchmarks.figures`. This paper's new geometry/Cp/flow-solution/H1-gallery figures (§4, §5.5): `.venv/bin/python -m cins.benchmarks.paper_figures`. Both re-run the underlying monolithic solves fresh rather than reading cached scalars, since neither `t7_naca2412/diagnostics.json` nor any `t8/*/result.json` stores the CST coefficient vectors or field distributions themselves (only scalar summaries; verified directly against the on-disk schema before writing `paper_figures.py`, not assumed). Individual T7/T8 cells: `.venv/bin/python experiments/run_t7.py` and `.venv/bin/python -m cins.benchmarks.runner` respectively. Tests: `python -m pytest tests/ -q`.

- **Gate-closure evidence.** [Sathish, 2026], `T3-closure.md`, `T4-T7-closure.md` record the six-condition gate-closure contract ([Sathish, 2026]) for each corresponding task, including adversarial-review findings and their resolution; [Sathish, 2026] through `ADR-0004` record the four binding architectural decisions this paper's Formulation section depends on (scipy shim, vendor NACA5 shim, forced-transition shim, realisability semantics).

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